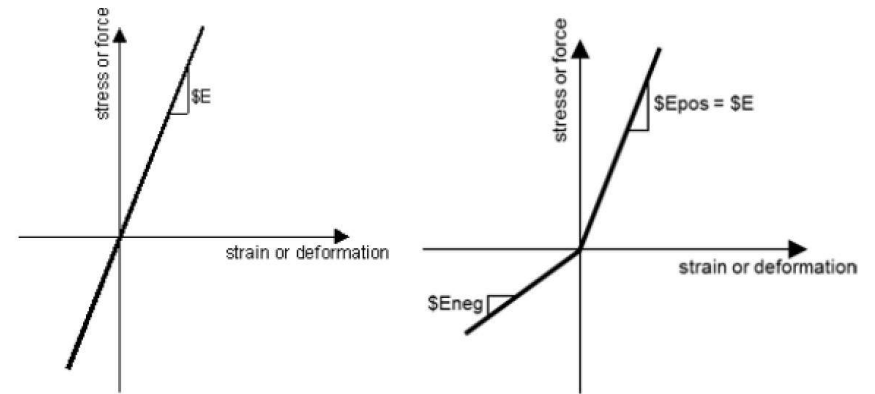
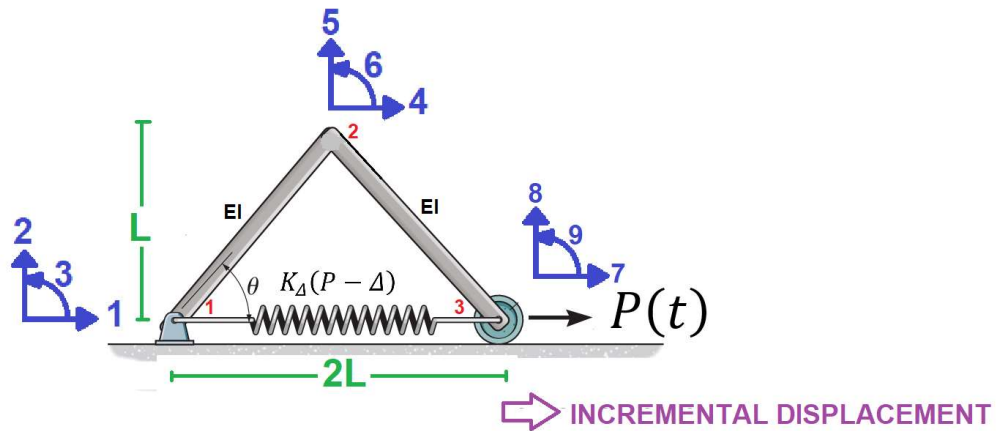


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A FRAME STRUCTURE: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME- HISTORY AND DYNAMIC TIME- HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



ELASTIC MATERIAL



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

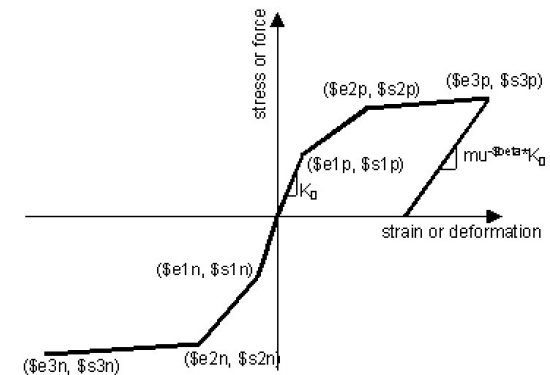
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



INELASTIC MATERIAL

STATIC ANALYSIS

Spyder (Python 3.12)

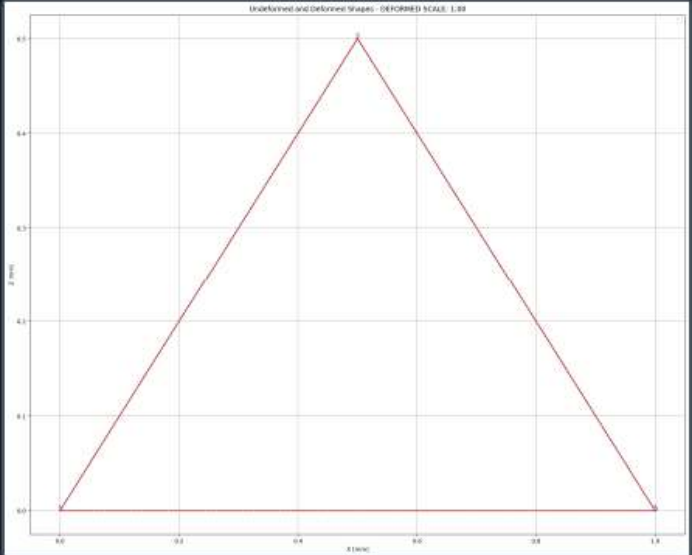
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\25_FRAME_TRUSS_8_ANA\P(t)_FRAME_TRUSS_8_ANA.py

P(t)_FRAME_TRUSS_8_ANA.py X

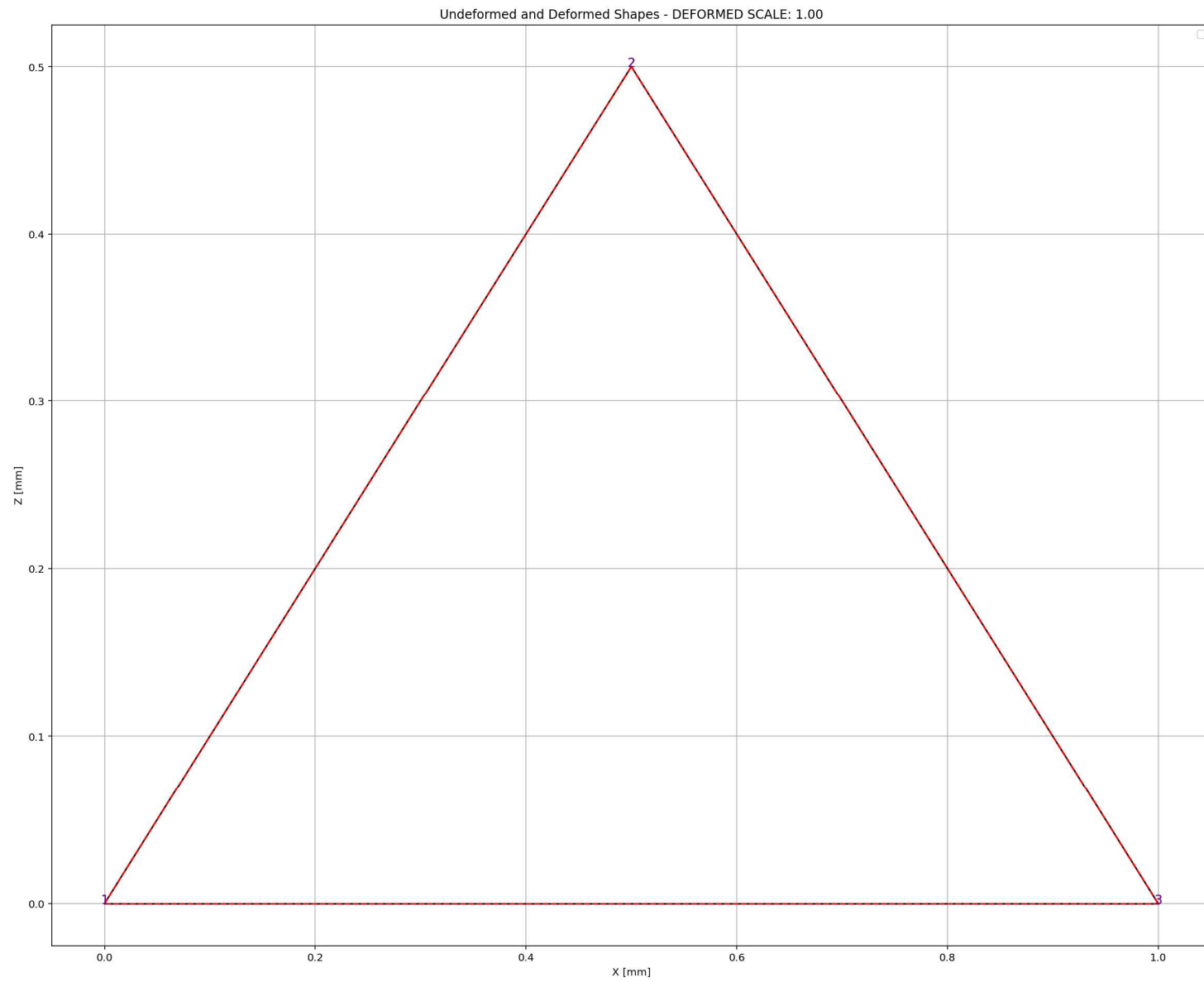
```
176 PERIOD_MIN, PERIOD_MAX = [], []
177
178 if ANAL_TYPE == 'PERIOD':
179     #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(2, 0.5*DR, DR, 0, 1)
180     PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(2, PLOT=True)
181     # Compute modal properties
182     ops.modalProperties("-print", "-file", f"SALAR_ModalReport_{ANAL_TYPE}.txt", "-unorm
183     return PERIODmin, PERIODmax
184
185 if ANAL_TYPE == 'STATIC':
186     ops.integrator('LoadControl', 1.0)
187     ops.analysis('Static')
188     OK = ops.analyze(1)
189     S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
190     ops.reactions()
191     reactionX.append(ops.nodeReaction(1, 1))          # BASE REACTION IN X DIR.
192     reactionY.append(ops.nodeReaction(1, 2))          # BASE REACTION IN Y DIR.
193     reactionZ.append(ops.nodeReaction(1, 3))          # BASE REACTION IN Z DIR.
194     dispX.append(ops.nodeDisp(center_node, 1))        # DISPLACEMENT IN X DIR.
195     dispY.append(ops.nodeDisp(center_node, 2))        # DISPLACEMENT IN Y DIR.
196     dispZ.append(ops.nodeDisp(center_node, 3))        # ROTATION IN Z DIR.
197     stiffnessX.append(np.abs(reactionX[-1]) / np.abs(dispX[-1]))
198     stiffnessY.append(np.abs(reactionY[-1]) / np.abs(dispY[-1]))
199     stiffnessZ.append(np.abs(reactionZ[-1]) / np.abs(dispZ[-1]))
200     print('\n\nSTATIC ANALYSIS DONE.\n\n')
201
202     DATA = (reactionX, dispX, velX, accelX,
203             reactionY, dispY, velY, accelY,
204             reactionZ, dispZ, velZ, accelZ,
205             stiffnessX, stiffnessY, stiffnessZ)
206
207     return DATA
208
209 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
```

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00



IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 121, Col 115 UTF-8 CRLF RW Mem 55%



PUSHOVER ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..EX\25_FRAME_TRUSS_8_ANA\P(t)_FRAME_TRUSS_8_ANA.py

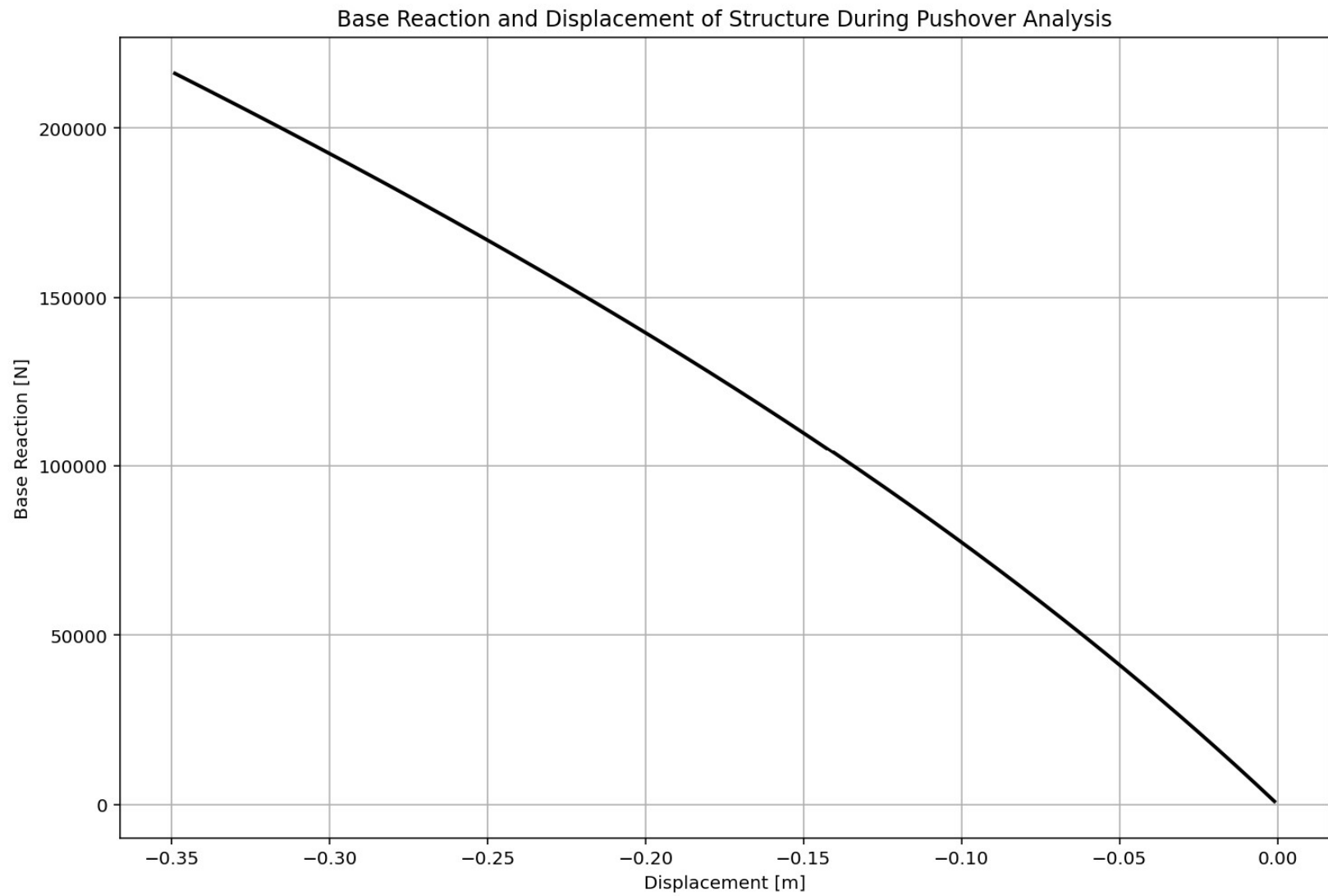
P(t)_FRAME_TRUSS_8_ANA.py X

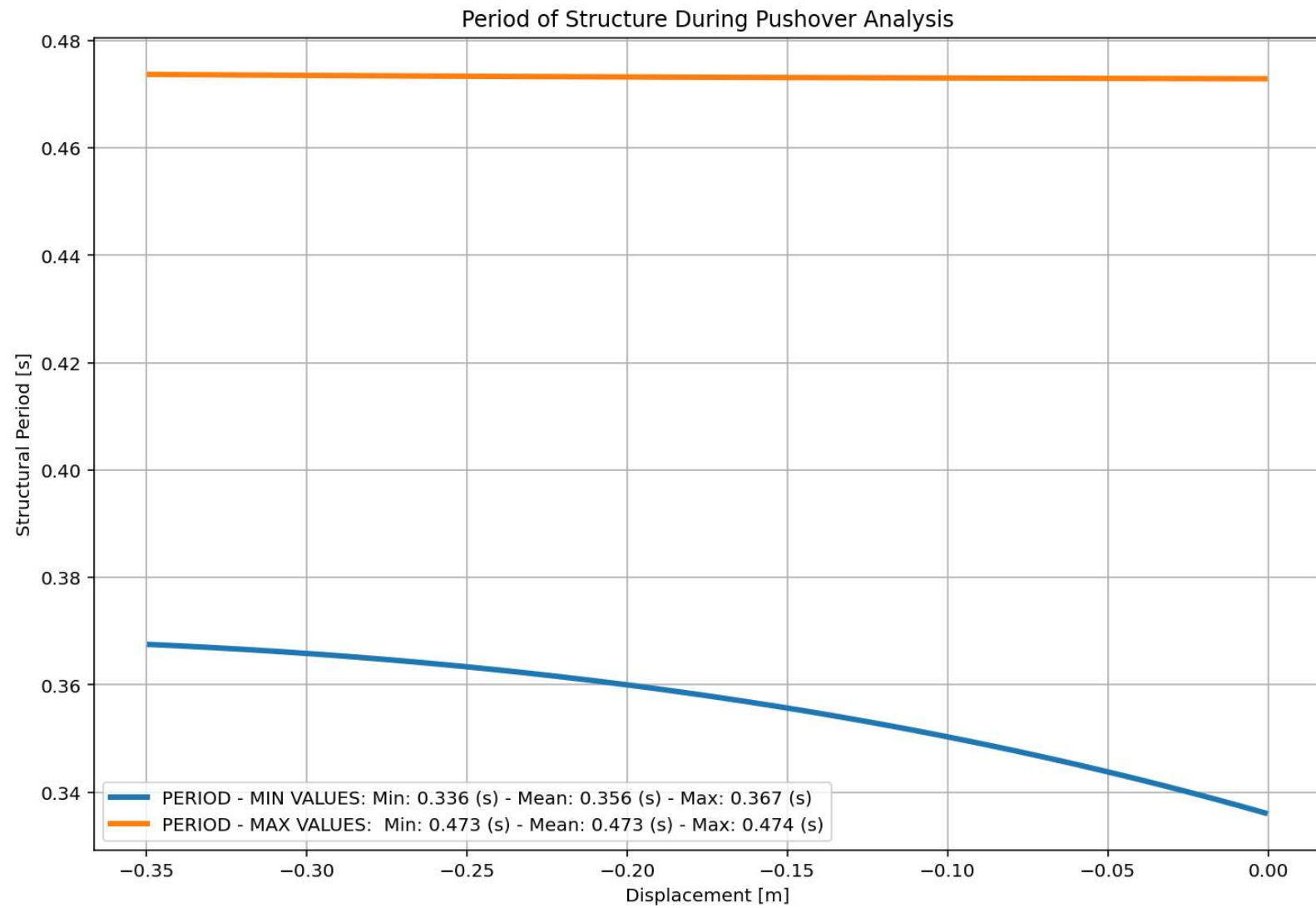
```
209 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
210     IDctrlDOF = 1 # 1: Horizontal Displcement - 2: Vertical Displcement
211     DINCR = -0.001 # [m] Incremental Vertical Displacement
212     DMAX = -0.35 # [m] Max. Displacement
213     ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
214     ops.analysis('Static')
215     Nsteps = int(np.abs(DMAX/ DINCR))
216     STEP = 0.0
217     for step in range(Nsteps):
218         OK = ops.analyze(1)
219         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
220         ops.reactions()
221         reactionX.append(ops.nodeReaction(1, 1)) # BASE REACTION IN X DIR.
222         reactionY.append(ops.nodeReaction(1, 2)) # BASE REACTION IN Y DIR.
223         reactionZ.append(ops.nodeReaction(1, 3)) # BASE REACTION IN Z DIR.
224         dispX.append(ops.nodeDisp(center_node, 1)) # DISPLACEMENT IN X DIR.
225         dispY.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT IN Y DIR.
226         dispZ.append(ops.nodeDisp(center_node, 3)) # ROTATION IN Z DIR.
227         velX.append(ops.nodeVel(center_node, 1)) # VELOCITY IN X DIR.
228         velY.append(ops.nodeVel(center_node, 2)) # VELOCITY IN Y DIR.
229         velZ.append(ops.nodeVel(center_node, 3)) # VELOCITY IN Z DIR.
230         accelX.append(ops.nodeAccel(center_node, 1)) # ACCELERATION IN X DIR.
231         accelY.append(ops.nodeAccel(center_node, 2)) # ACCELERATION IN Y DIR.
232         accelZ.append(ops.nodeAccel(center_node, 3)) # ACCELERATION IN Z DIR.
233         stiffnessX.append(np.abs(reactionX[-1]) / np.abs(dispX[-1]))
234         stiffnessY.append(np.abs(reactionY[-1]) / np.abs(dispY[-1]))
235         stiffnessZ.append(np.abs(reactionZ[-1]) / np.abs(dispZ[-1]))
236         # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
237         #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(2, 0.5*DR, DR, 0, 1)
238         PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(2, PLOT=True)
239         PERIOD_MIN.append(PERIODmin)
240         PERIOD_MAX.append(PERIODmax)
241         STEP += 1
242         #print(STEP, disp[-1], reaction[-1])
```

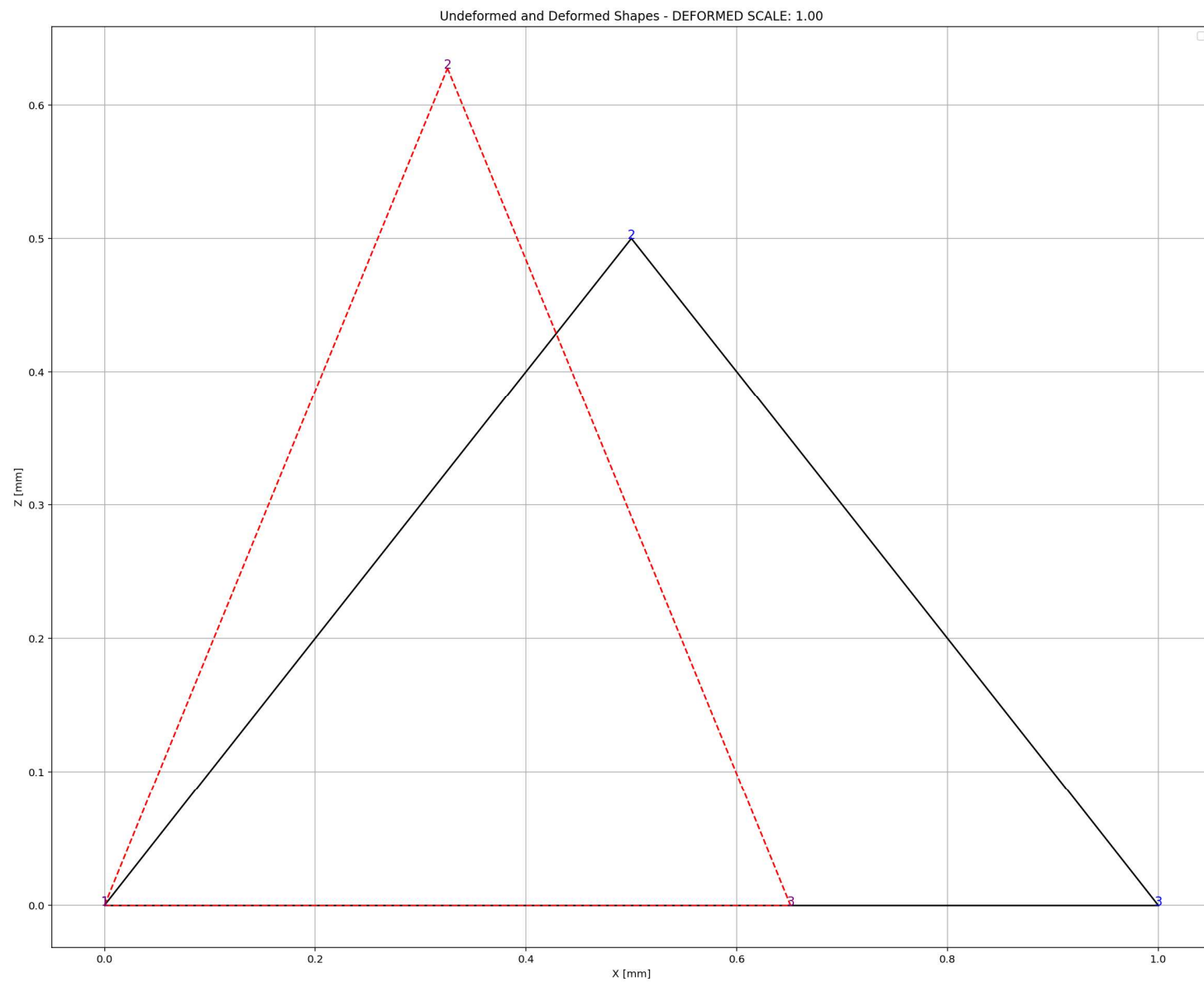
Base Reaction and Displacement of Structure During Pushover Analysis

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 257, Col 18 UTF-8 CRLF RW Mem 47%







CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

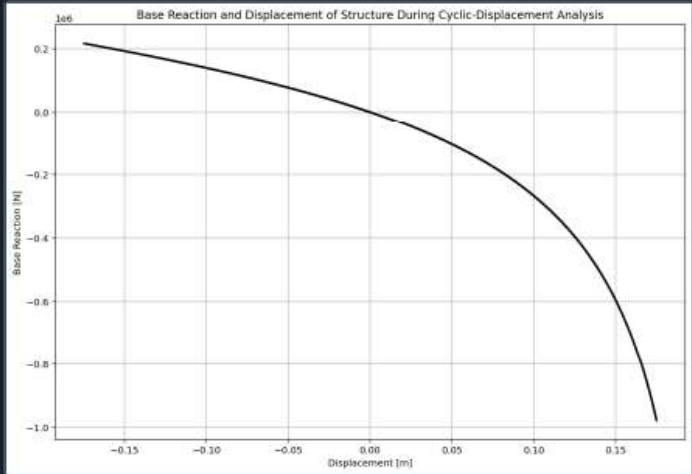
File Edit Search Source Run Debug Consoles Projects Tools View Help

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P(t)_FRAME_TRUSS_8_ANA.py

```
255 if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
256     IDctrlDOF = 1 # 1: Horizontal Displacement - 2: Vertical Displacement
257     DMAX = -0.35 # [m] Max. Displacement
258     n_points = 10000
259     # 4. CYCLIC DISPLACEMENT PROTOCOL
260     # Key strain points (same logic as your protocol)
261     key_disp = np.array([
262         0.0,
263         0.1*DMAX, -0.1*DMAX,
264         0.5*DMAX, -0.5*DMAX,
265         0.8*DMAX, -0.8*DMAX,
266         DMAX, -DMAX,
267         0.1*DMAX, -0.1*DMAX,
268         0.5*DMAX, -0.5*DMAX,
269         0.8*DMAX, -0.8*DMAX,
270         DMAX, -DMAX,
271         0.0
272     ])
273
274     # Generate 1000-point displacement protocol
275     disp_protocol = np.interp(
276         np.linspace(0, len(key_disp) - 1, n_points),
277         np.arange(len(key_disp)),
278         key_disp
279     )
280
281     ops.analysis('Static')
282     STEP = 0.0
283     for target_disp in disp_protocol:
284         current_disp = ops.nodeDisp(center_node, 1) # DISPLACEMENT APPLIED IN MIDDLE N
285         dU = target_disp - current_disp
286         ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
287         OK = ops.analyze(1)
288         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
```

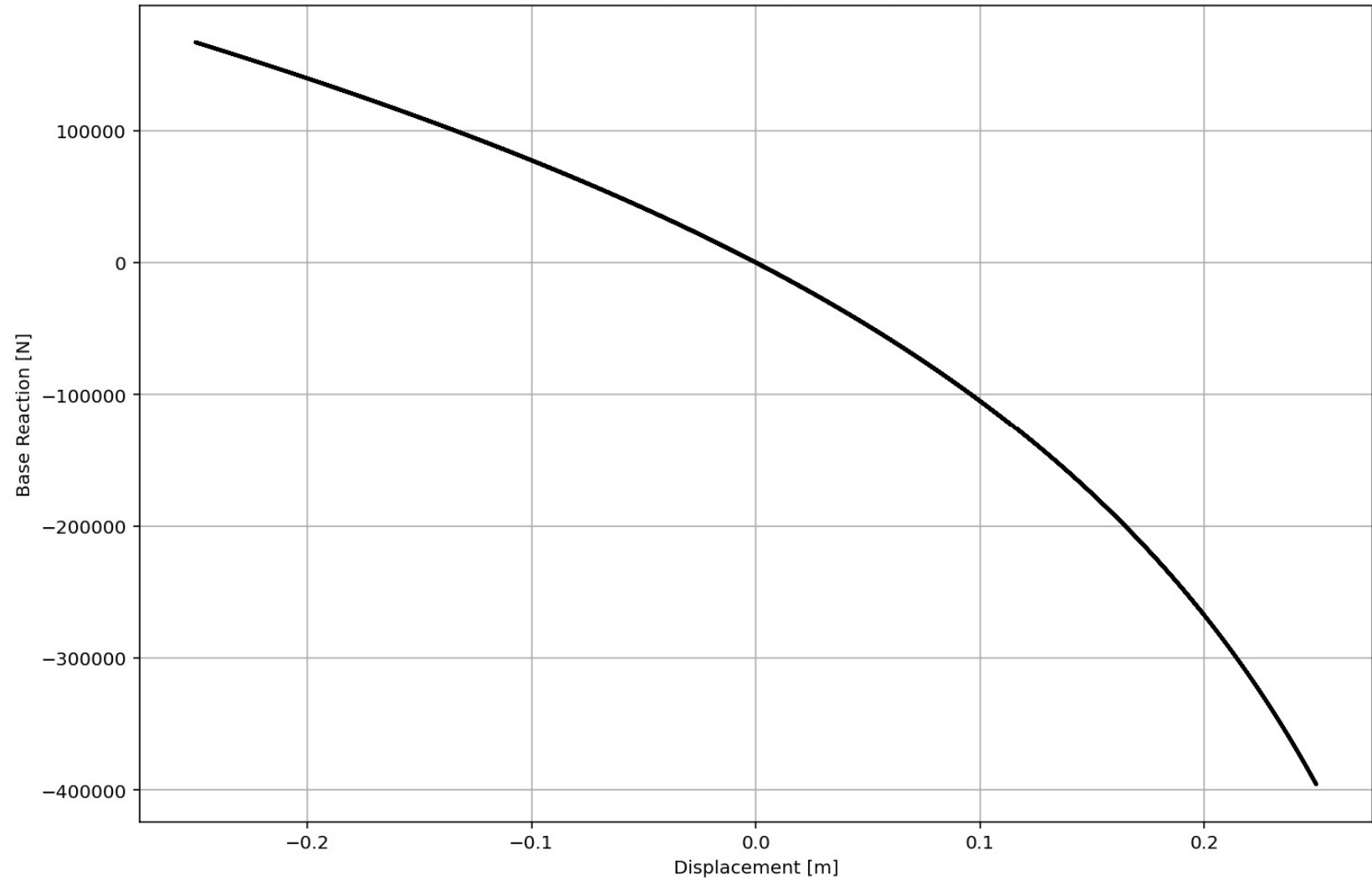
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis



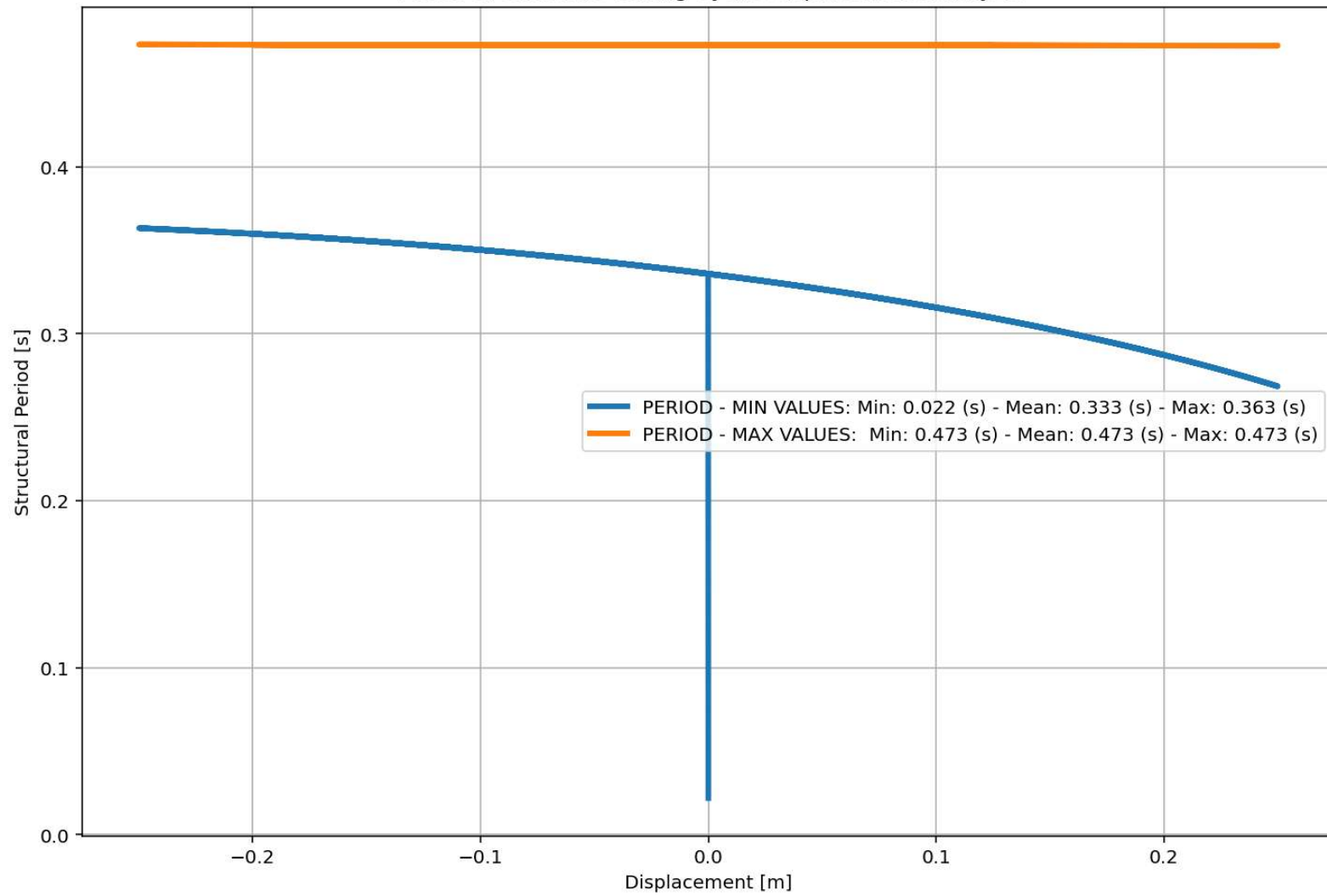
IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 257, Col 18 UTF-8 CRLF RW Mem 47%

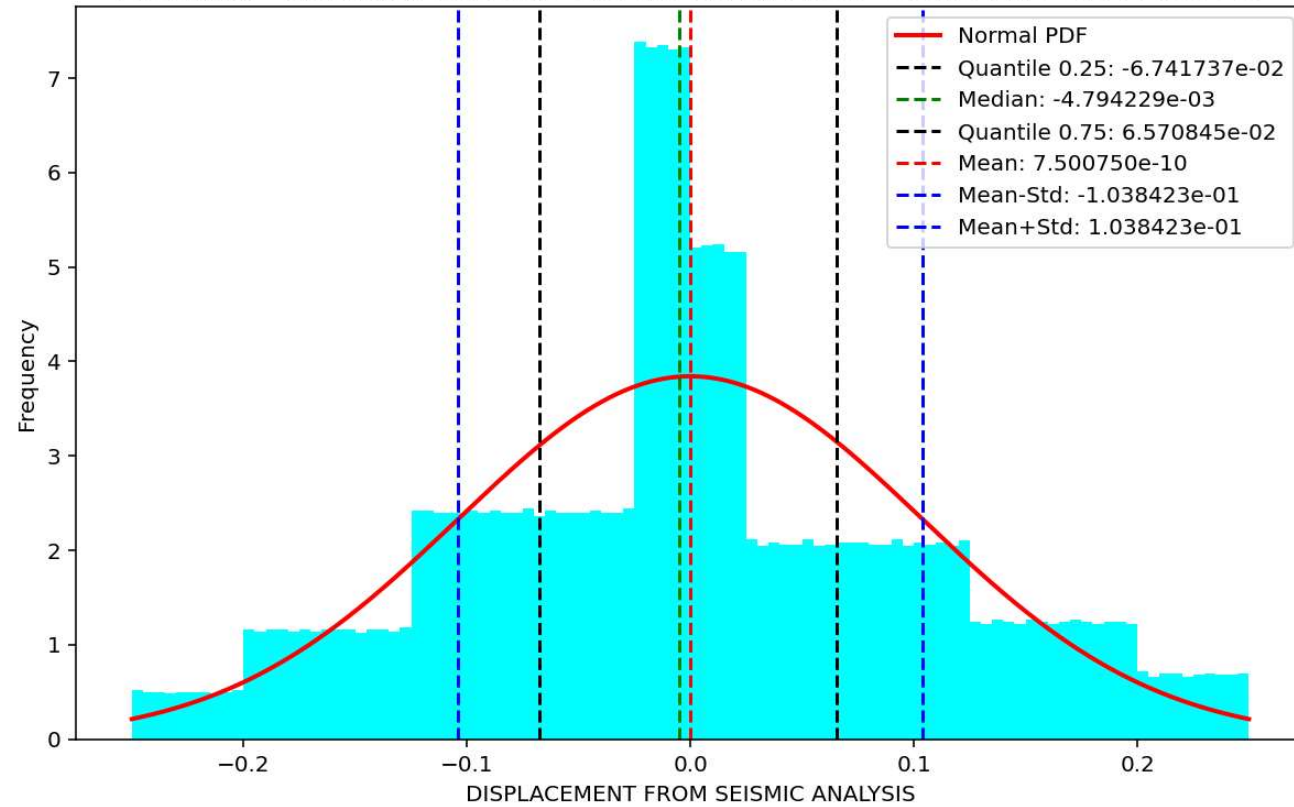
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

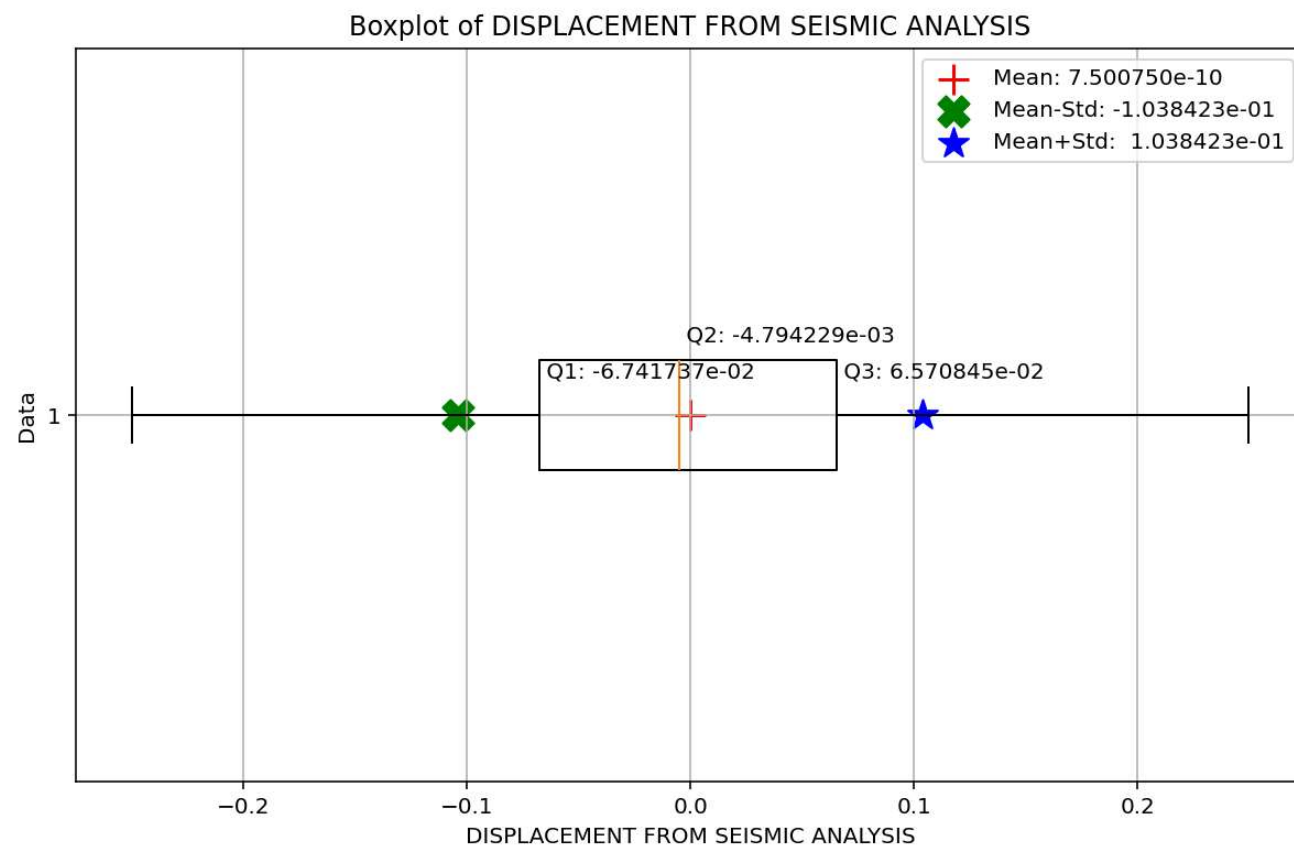


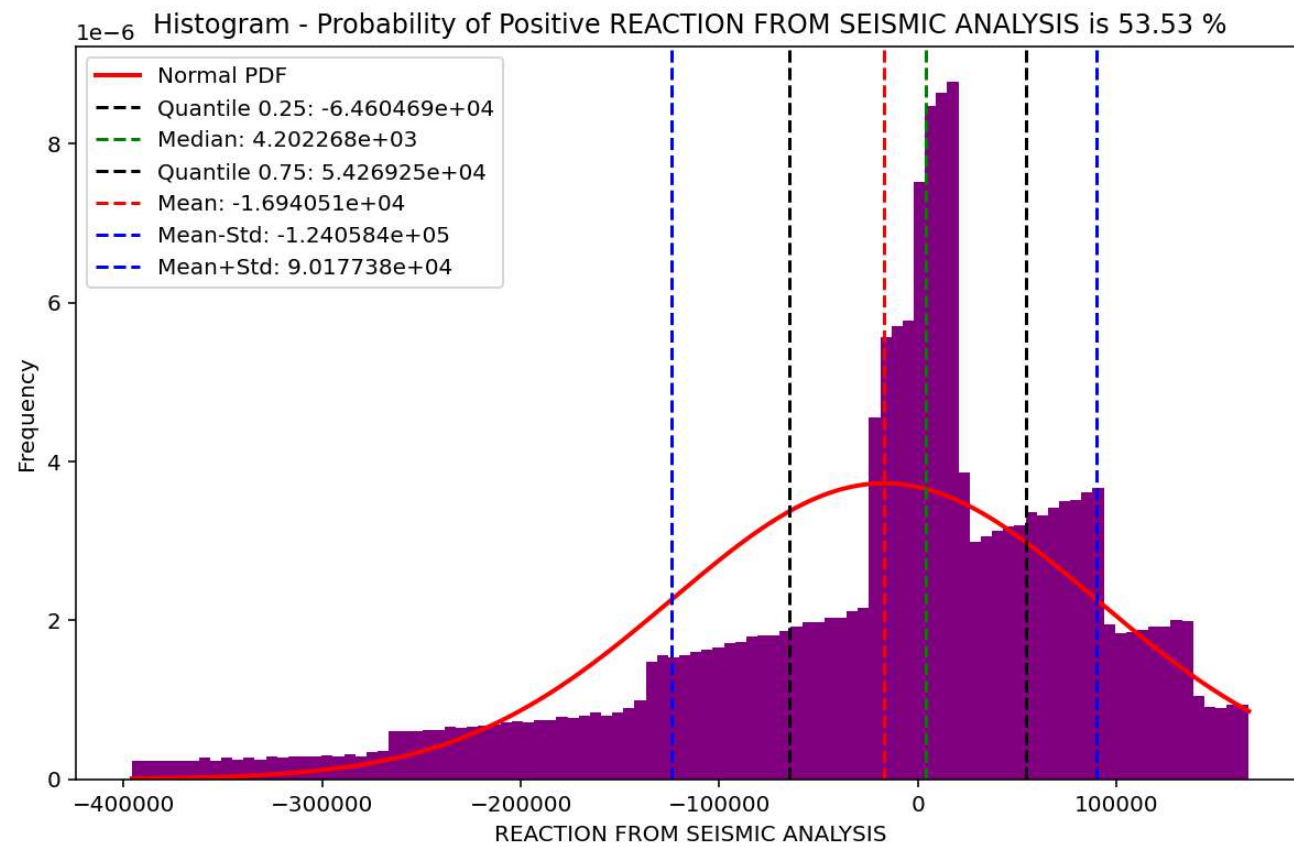
Period of Structure During Cyclic Displacement Analysis

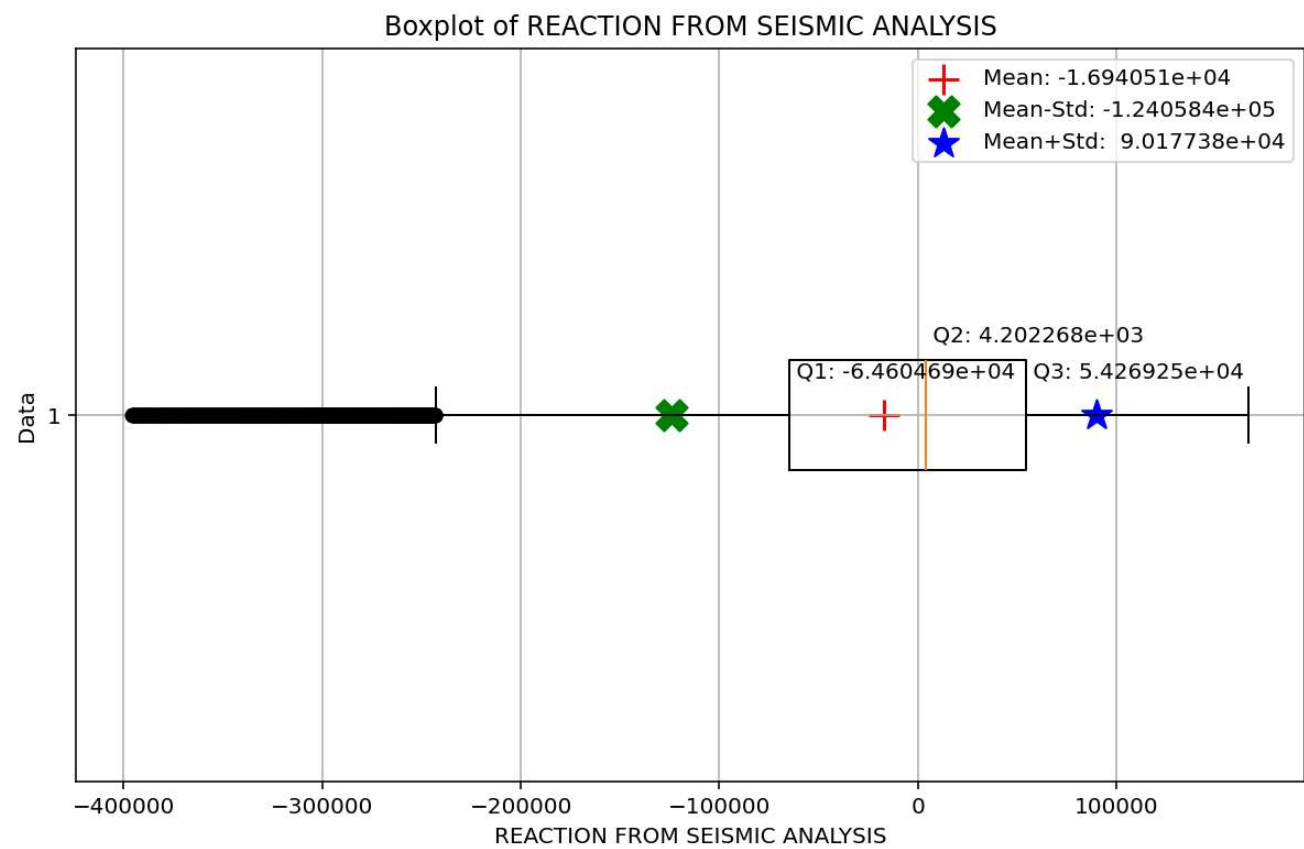


Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 46.46 %

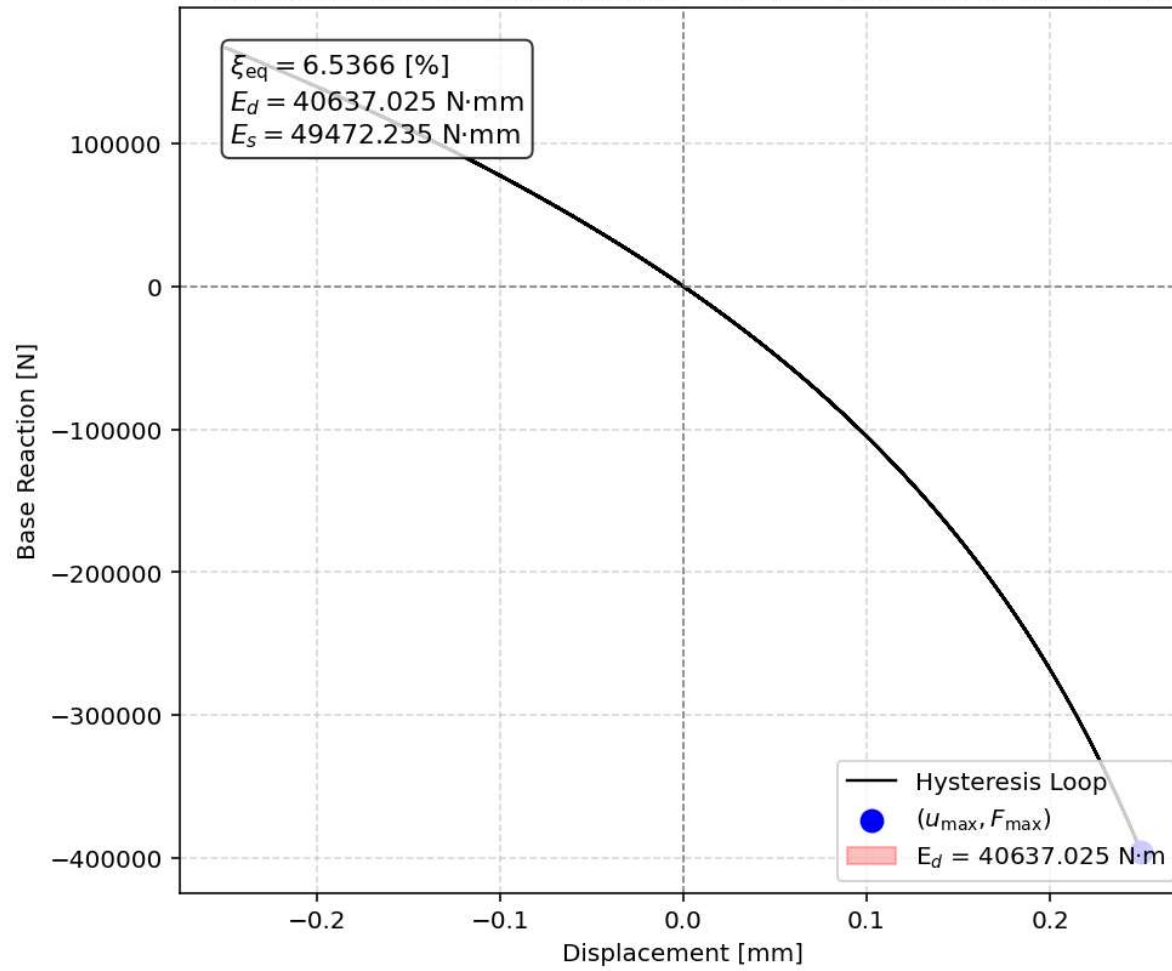


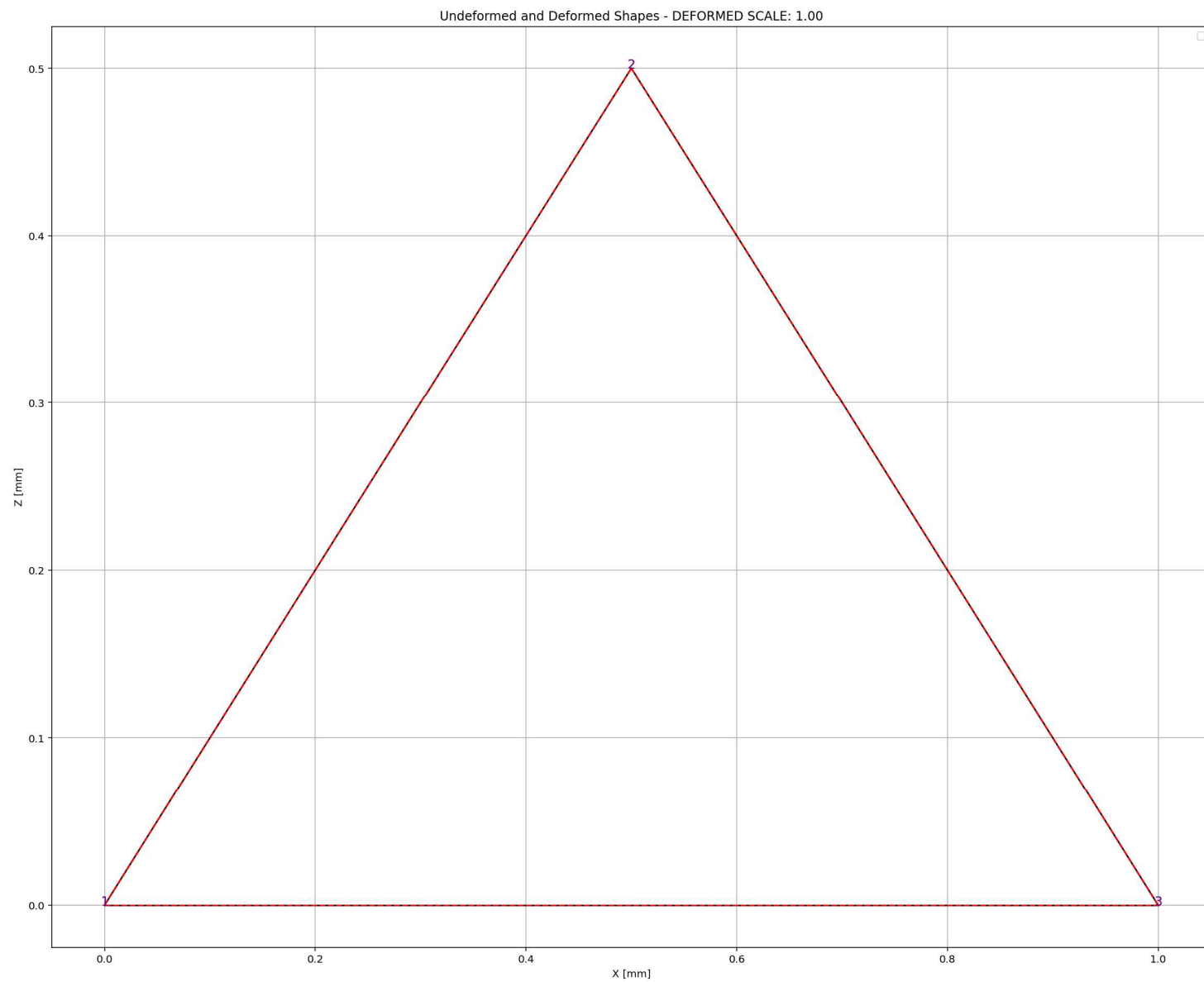






Equivalent viscous damping ratio - Cyclic-Displacement Hysteresis





STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

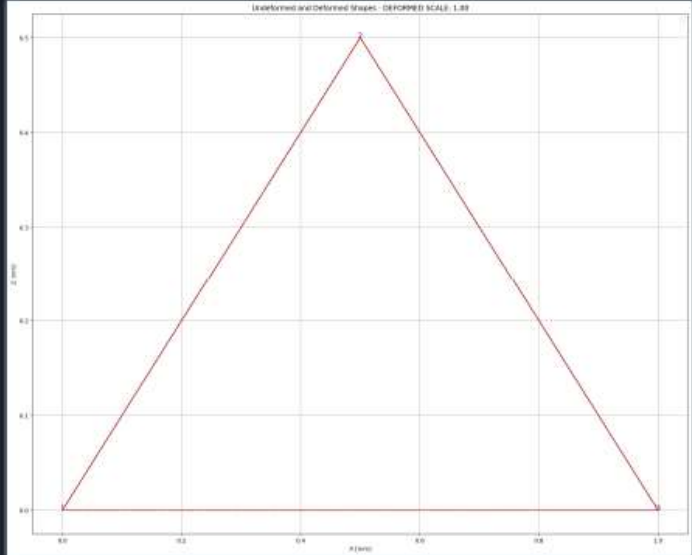
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\25_FRAME_TRUSS_8_ANA\P(t)_FRAME_TRUSS_8_ANA.py

P(t)_FRAME_TRUSS_8_ANA.py

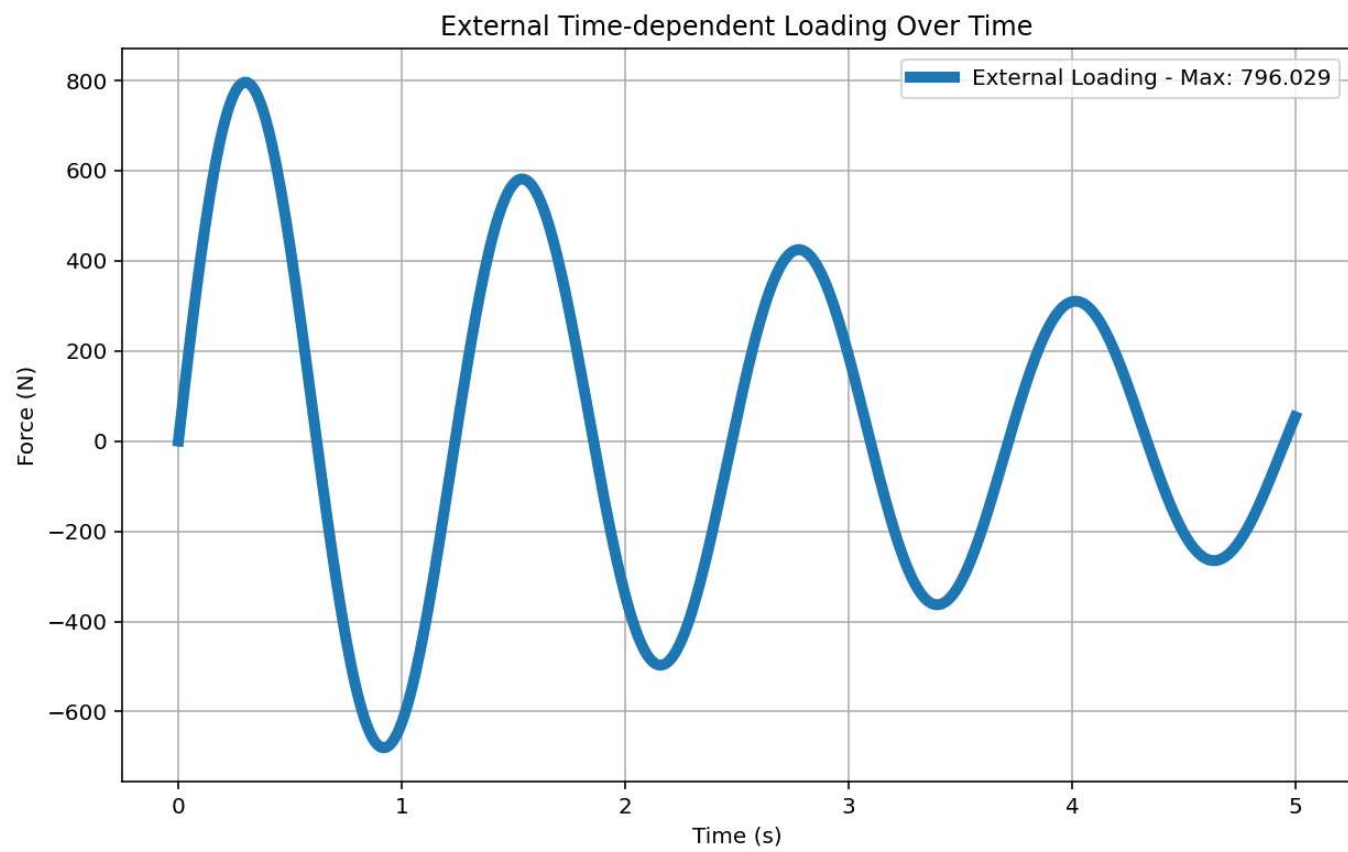
```
318 if ANAL_TYPE == 'STATIC EXTERNAL TIME-DEPENDENT LOADING': # STATIC TIME-HISTORY ANALYSIS
319     ### DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
320     # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
321     duration = 20.0          # [s] Analysis duration
322     dt = 0.01               # [s] Time step
323     DT = dt                 # [s] Time step
324     DT_time = 5.0           # [s] Total external Load Analysis Durations [*****]
325     force_amplitude = 960.0 # [N] Amplitude Force
326     omega_DT = 5.0715       # [rad/s] Natural angular frequency
327     time_steps = int(duration/dt)
328
329     # Check function
330     def CHECK_FUN(DT_time ,duration):
331         if DT_time > duration:
332             print('\n\nAnalysis Duration Must be greater than External Load Duration\n')
333             exit()
334         return -1
335
336     CHECK_FUN(DT_time ,duration)
337     def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin(omega*t)
338         import numpy as np
339         import matplotlib.pyplot as plt
340         # External Load Durations
341         num_steps = int(DT_time / DT)
342         load_time = np.linspace(0, DT_time, num_steps)
343         target_frequency = 1.0 * omega_DT # Target excitation frequency
344         DT_load = force_amplitude * np.sin(target_frequency * load_time)
345         # Plot External Time-dependent Loading
346         plt.figure(figsize=(10, 6))
347         plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):.2f}')
348         plt.title('External Time-dependent Loading Over Time')
349         plt.xlabel('Time (s)')
350         plt.ylabel('Force (N)')
351         plt.grid(True)
```

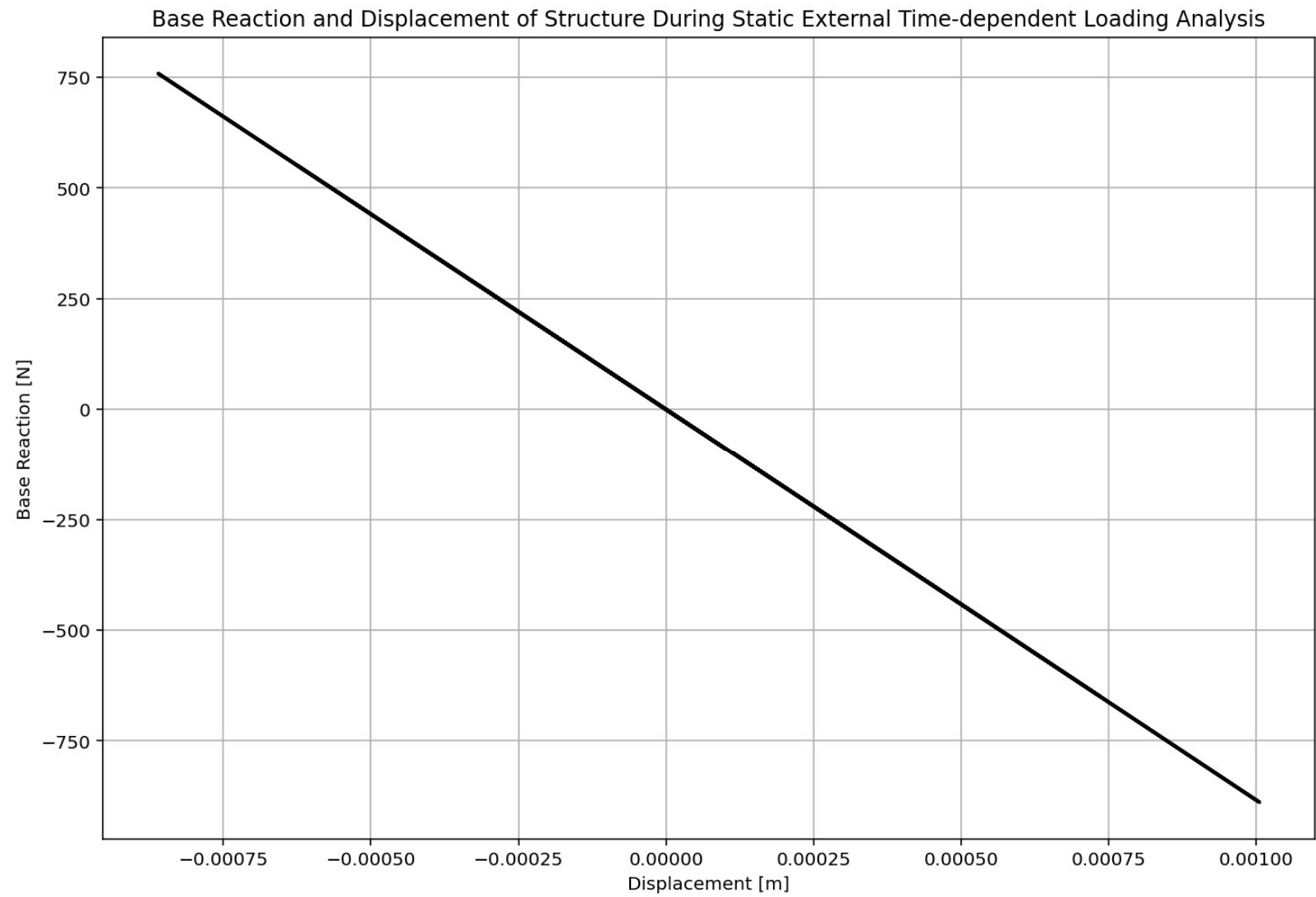
19 %



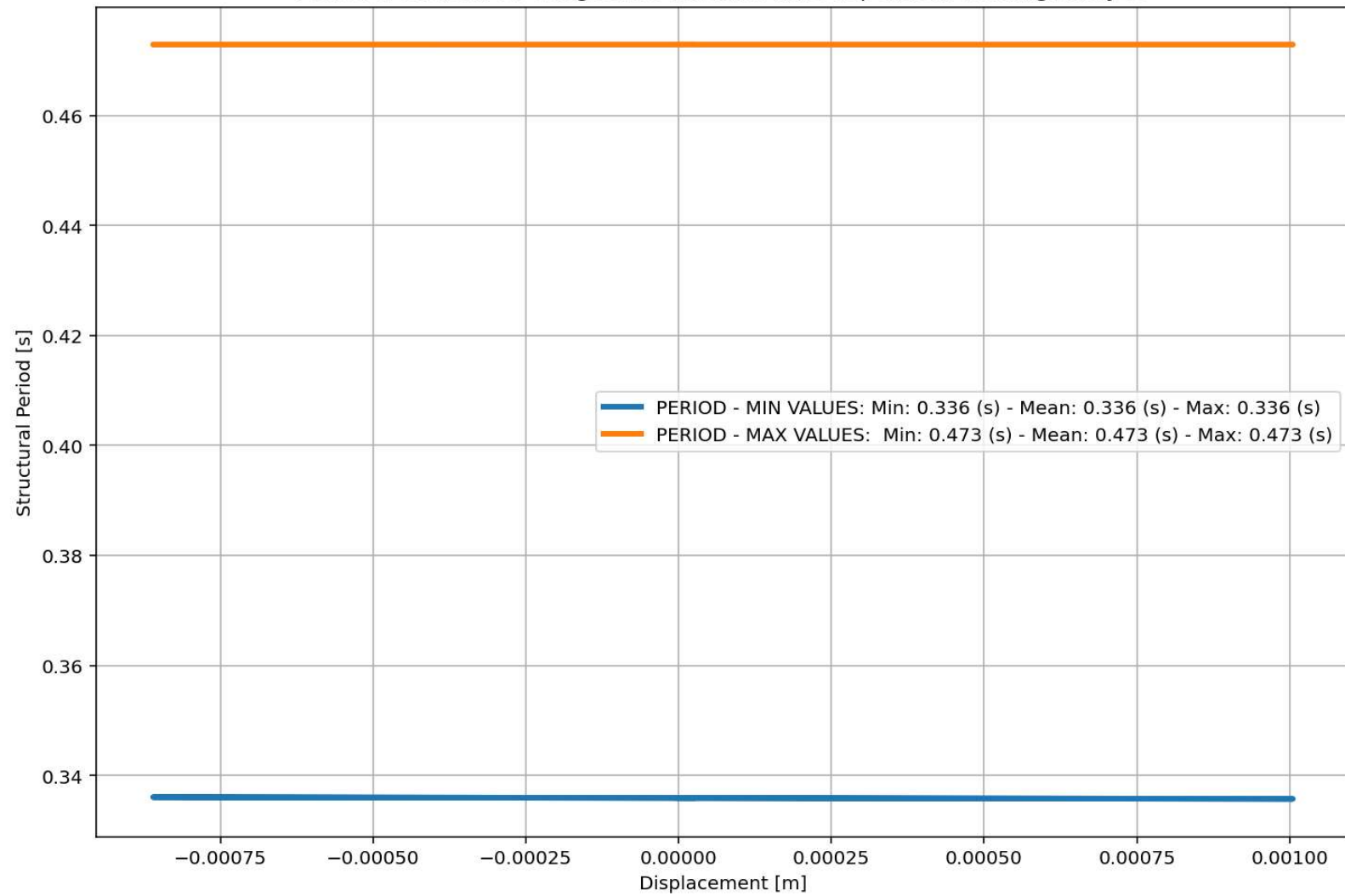
IPython Console Files Help Variable Explorer Debugger Plots History

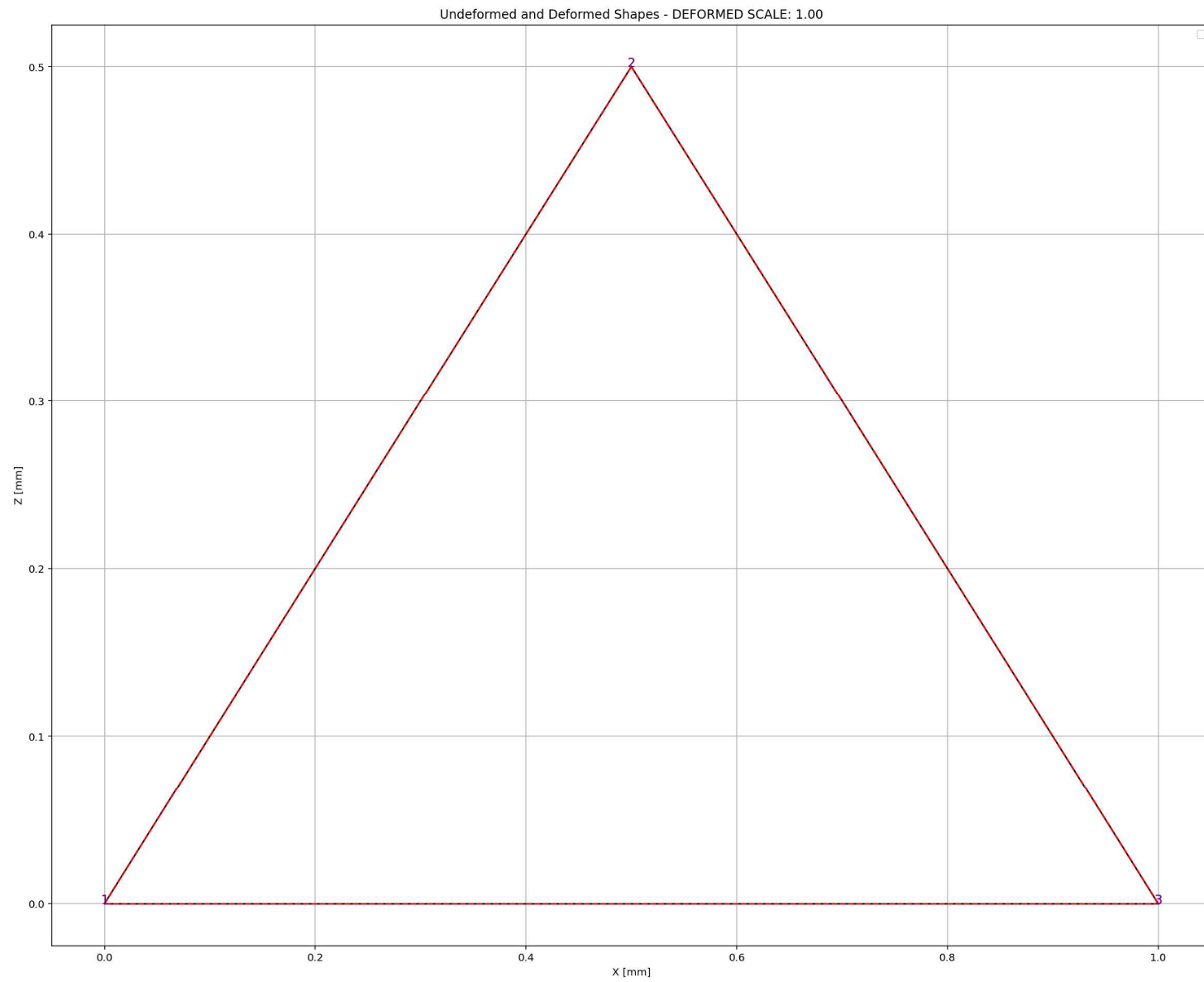
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 121, Col 115 UTF-8 CRLF RW Mem 56%





Period of Structure During Static External Time-dependent Loading Analysis





DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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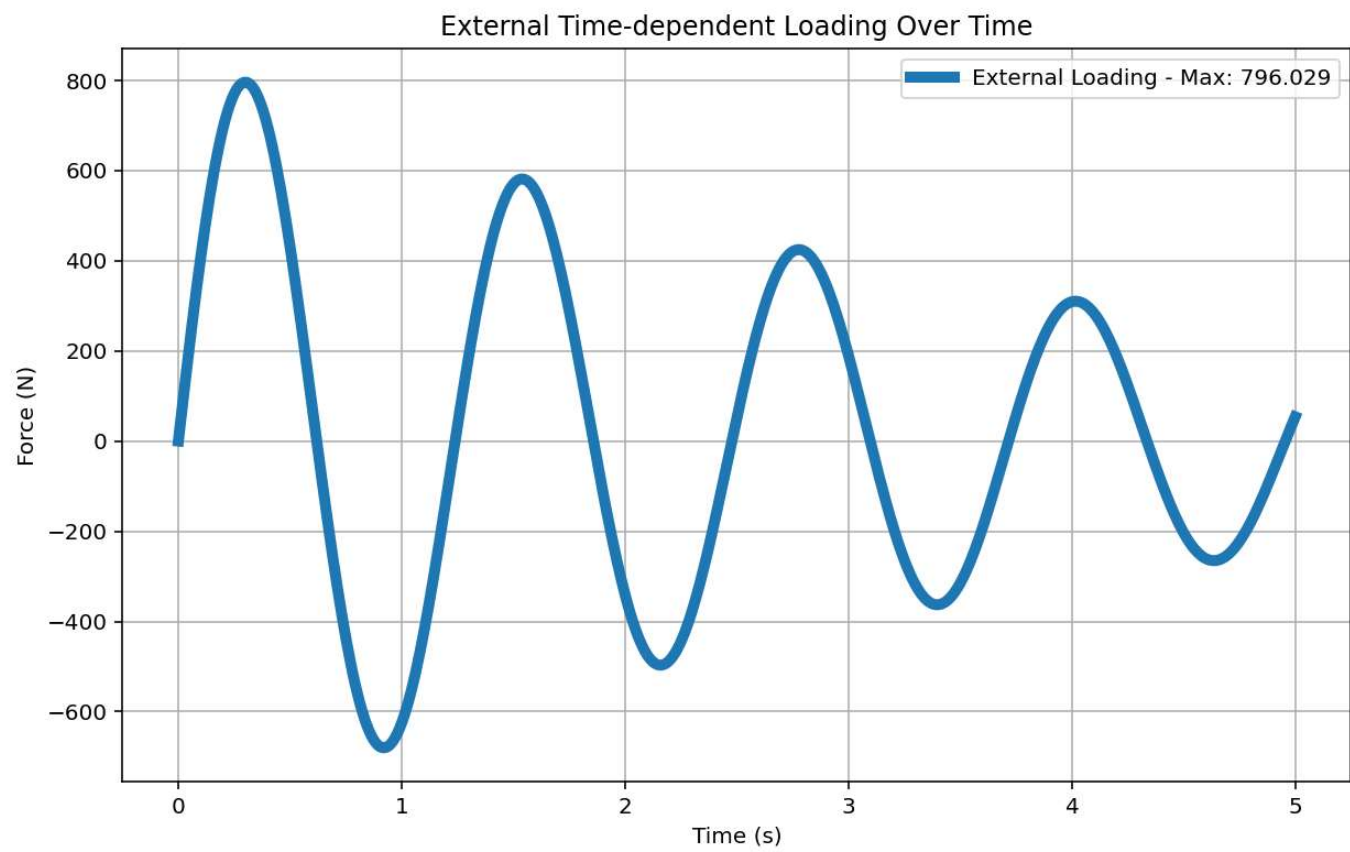
P(t)_FRAME_TRUSS_8_ANA.py X

```
429 if ANAL TYPE == 'DYNAMIC EXTERNAL TIME-DEPENDENT LOADING': # DYNAMIC TIME-HISTORY ANAL
430     %% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
431     duration = 20.0           # [s] Analysis duration
432     dt = 0.01                # [s] Time step
433     DT = dt                  # [s] Time step
434     DT_time = 5.0            # [s] Total external Load Analysis Durations [*****]
435     force_amplitude = 960.0   # [N] Amplitude Force
436     omega_DT = 5.0715         # [rad/s] Natural angular frequency
437     DR = 0.03                # Damping Ratio
438
439     # Check function
440     def CHECK_FUN(DT_time ,duration):
441         if DT_time > duration:
442             print('\n\nAnalysis Duration Must be greater than External Load Duration\n')
443             exit()
444         return -1
445
446     CHECK_FUN(DT_time ,duration)
447     def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 s
448         import numpy as np
449         import matplotlib.pyplot as plt
450         # External Load Durations
451         num_steps = int(DT_time / DT)
452         load_time = np.linspace(0, DT_time, num_steps)
453         target_frequency = 1.0 * omega_DT # Target excitation frequency
454         DT_load = force_amplitude * np.sin(target_frequency * load_time)
455         # Plot External Time-dependent Loading
456         plt.figure(figsize=(10, 6))
457         plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_Load)}:
458         plt.title('External Time-dependent Loading Over Time')
459         plt.xlabel('Time (s)')
460         plt.ylabel('Force (N)')
461         plt.grid(True)
462         plt.legend()
```

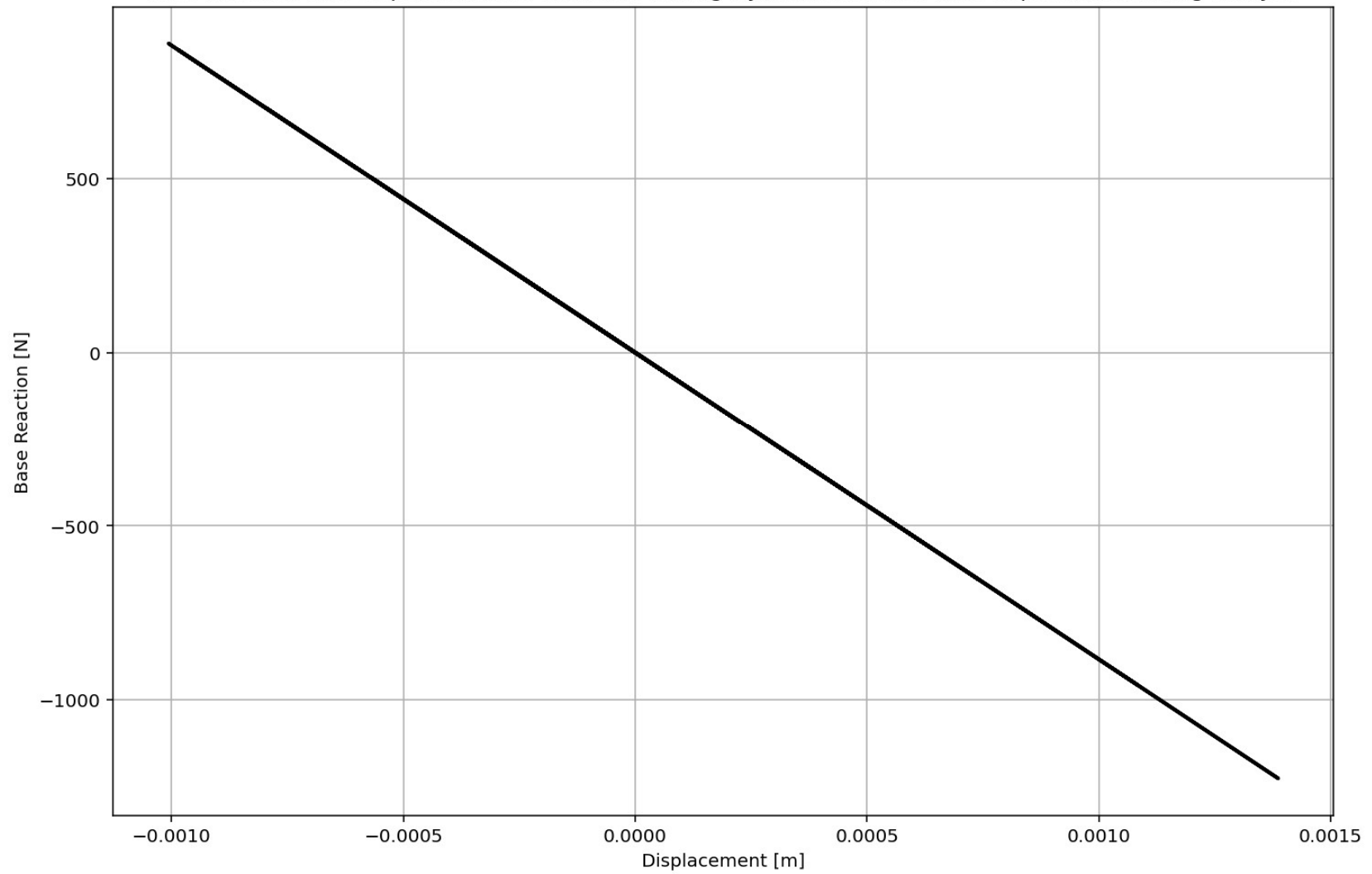
21 %

IPython Console Files Help Variable Explorer Debugger Plots History

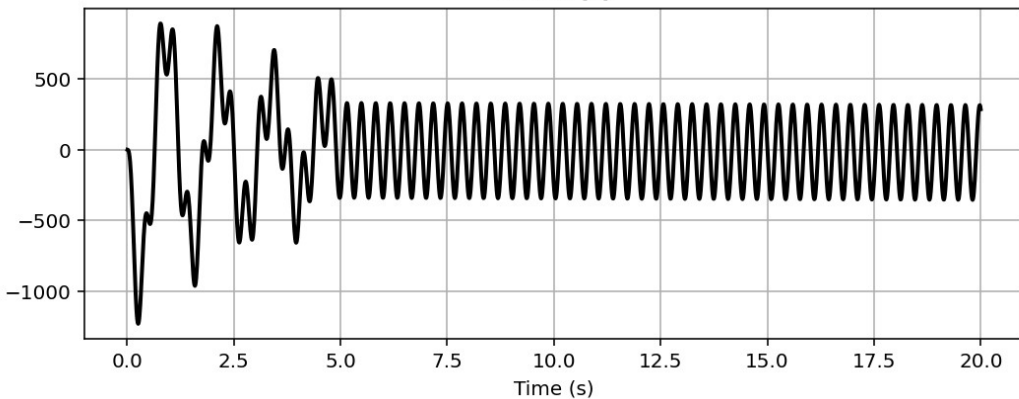
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 121, Col 115 UTF-8 CRLF RW Mem 56%



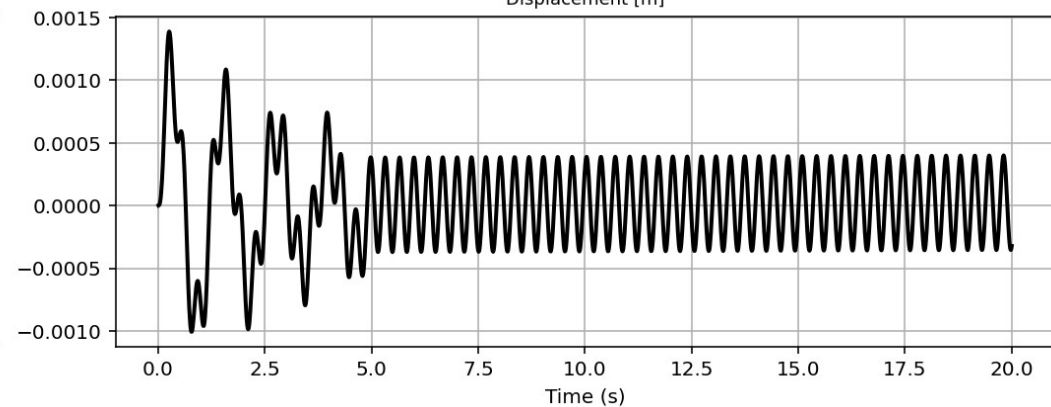
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



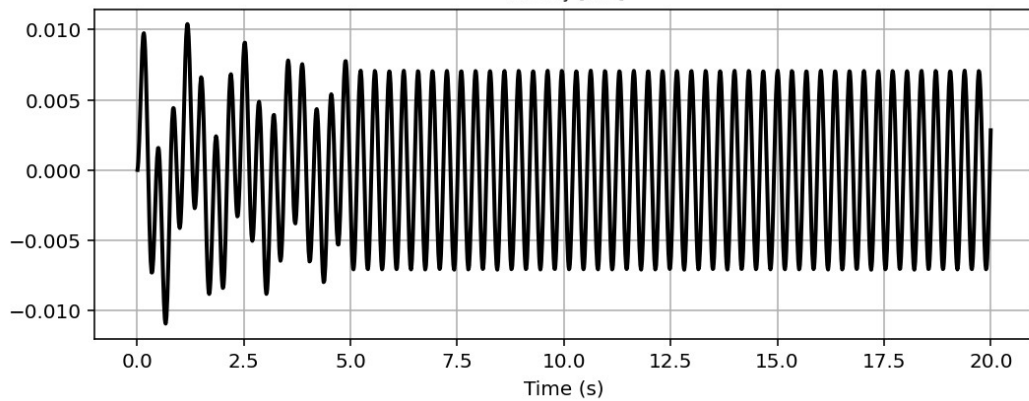
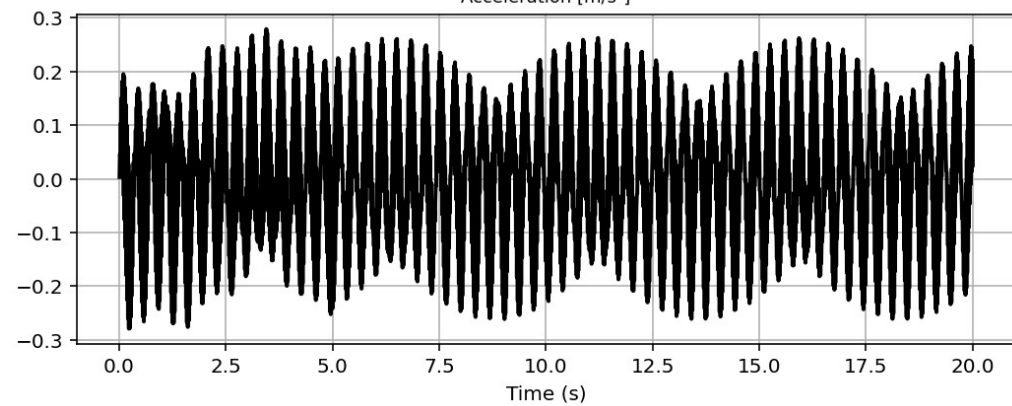
Reaction [N]



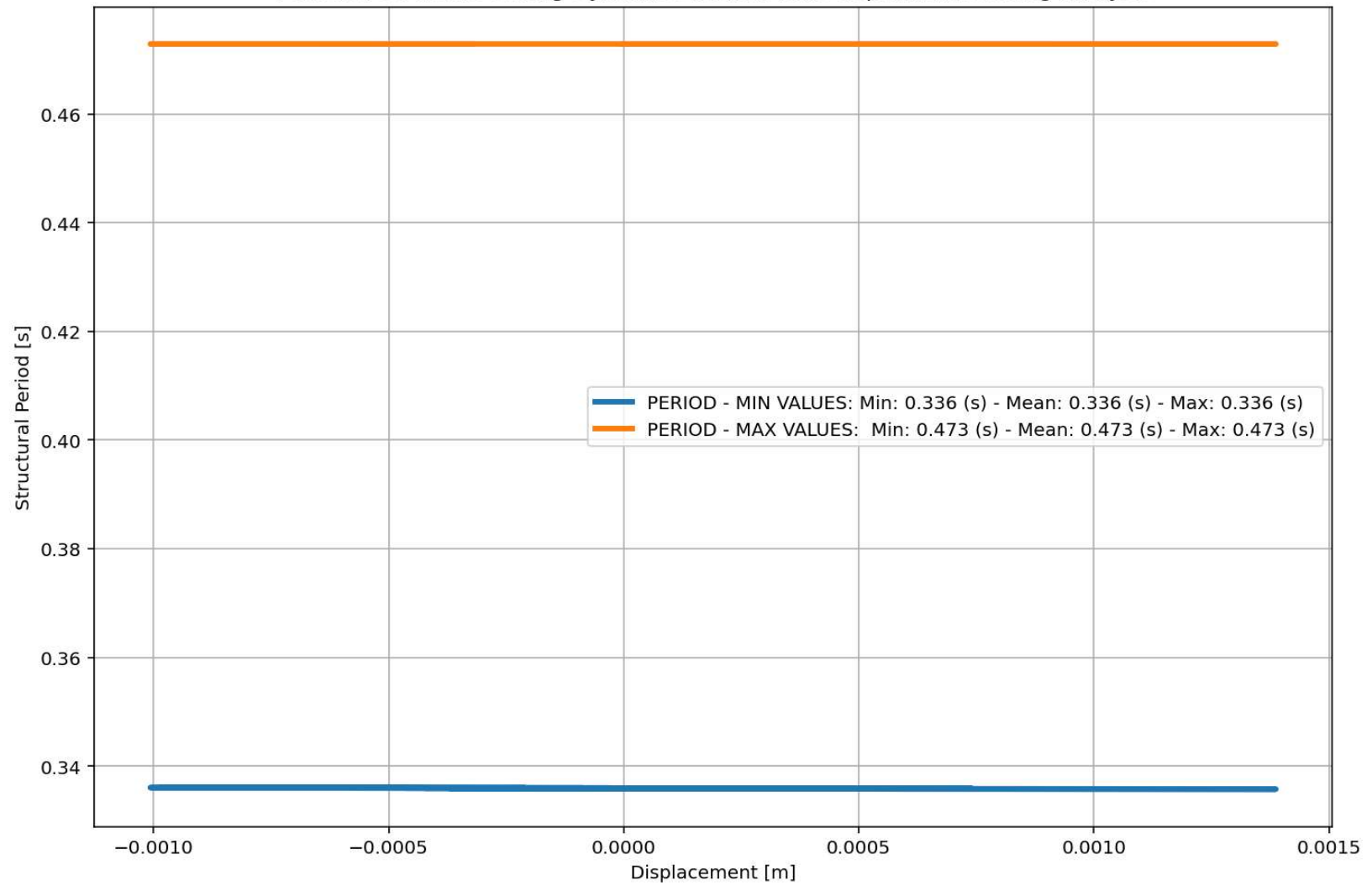
Displacement [m]

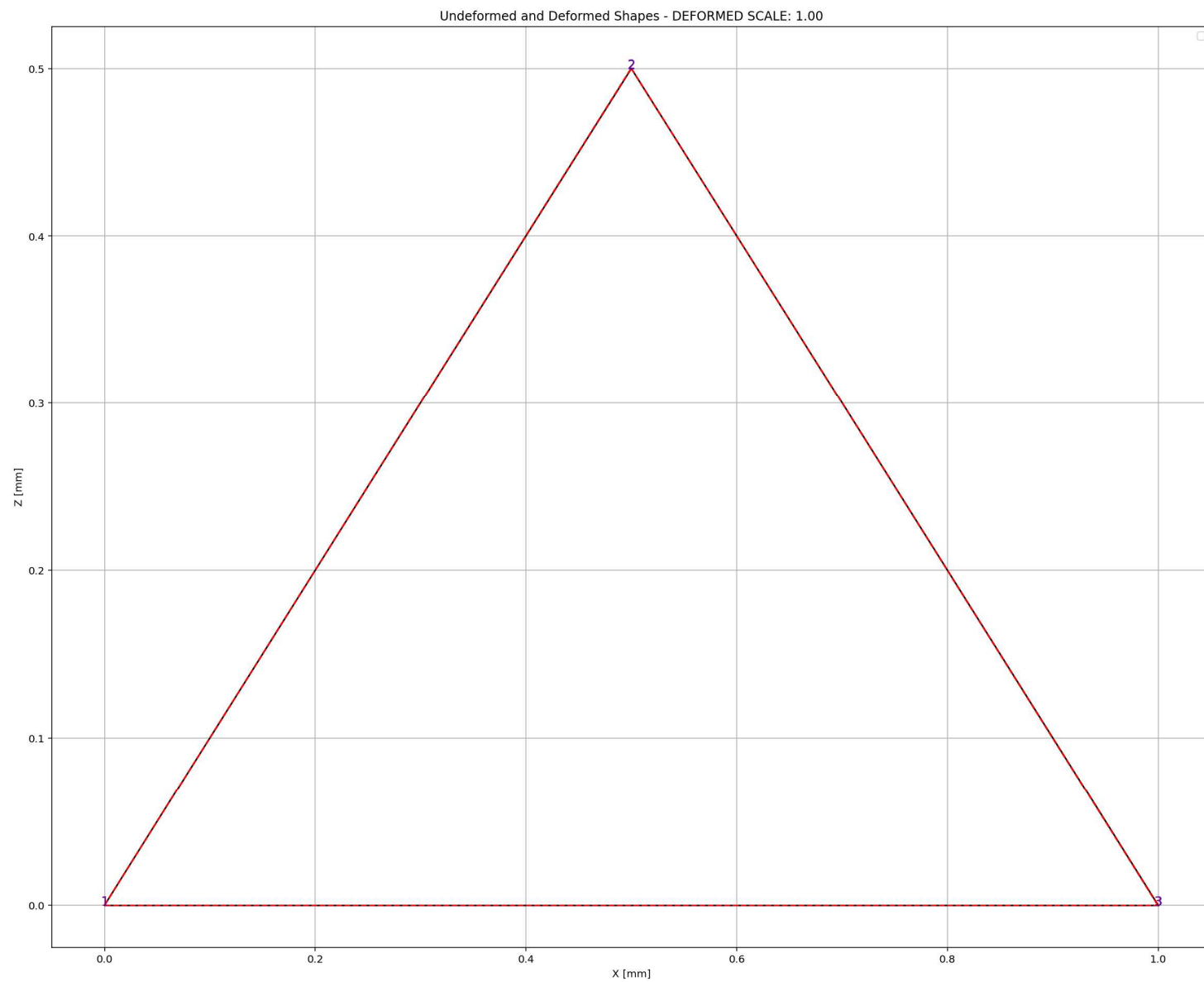


Velocity [m/s]

Acceleration [m/s²]

Period of Structure During Dynamic External Time-dependent Loading Analysis





FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

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P(t)_FRAME_TRUSS_8_ANA.py X

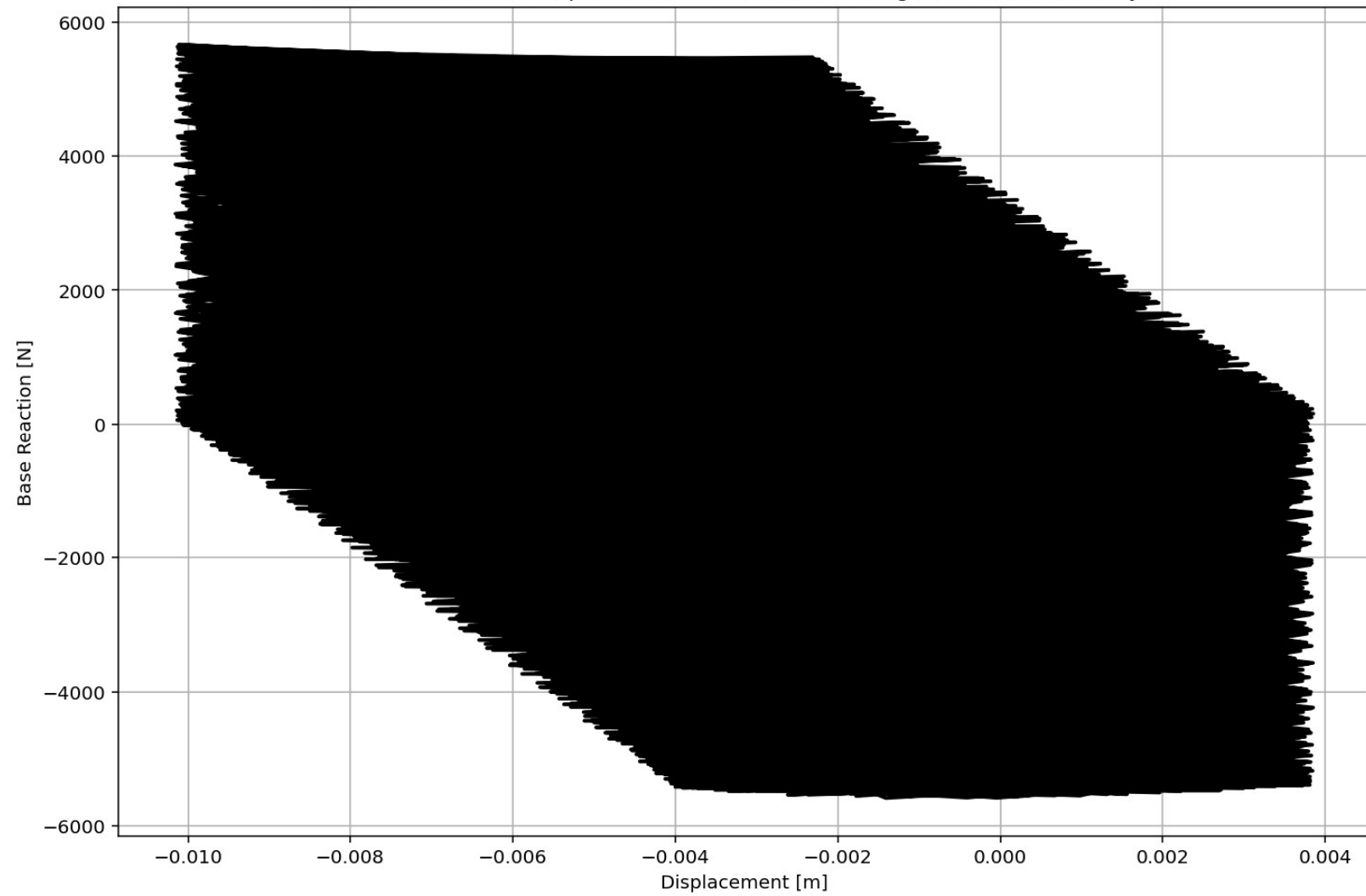
```
559 if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
560     ### DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
561     u0 = -0.01 # [m] Initial displacement
562     v0 = 0.150 # [m/s] Initial velocity
563     a0 = 0.065 # [m/s^2] Initial acceleration
564     IU = True # Free Vibration with Initial Displacement
565     IV = True # Free Vibration with Initial Velocity
566     IA = True # Free Vibration with Initial Acceleration
567     duration = 20.0 # [s] Analysis duration
568     dt = 0.001 # [s] Time step
569     DR = 0.03 # Damping Ratio
570
571     if IU == True:
572         # Define initial displacment
573         ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: https://openseespy
574     if IV == True:
575         # Define initial velocity
576         ops.setNodeVel(center_node, 1, v0, '-commit') # INFO LINK: https://openseespy
577     if IA == True:
578         # Define initial acceleration
579         ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: https://openseespy
580
581     ops.constraints('Plain')
582     ops.numberer('Plain')
583     ops.system('BandGeneral')
584     ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://openseespydoc.readthedocs.io/en/latest
585     #ops.integrator('CentralDifference') # JUST FOR LINEAR ANALYSIS - INFO LINK: https://openseespydoc.readthedocs.io/en/latest
586     alpha=0.5; beta=0.25;
587     ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthedocs.io/en/latest
588     #alpha=2/3; gamma=1.5-alpha; gamma=1.5-alpha; beta=(2-alpha)**2/4;
589     #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.readthedocs.io/en/latest
590     ops.algorithm('Newton') # INFO LINK: https://openseespydoc.readthedocs.io/en/latest
591     ops.analysis('Transient') # INFO LINK: https://openseespydoc.readthedocs.io/en/latest
592
```

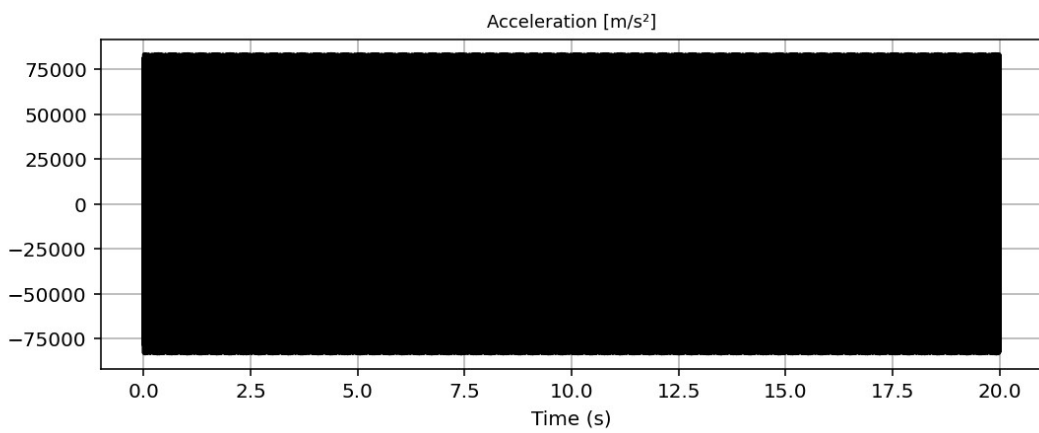
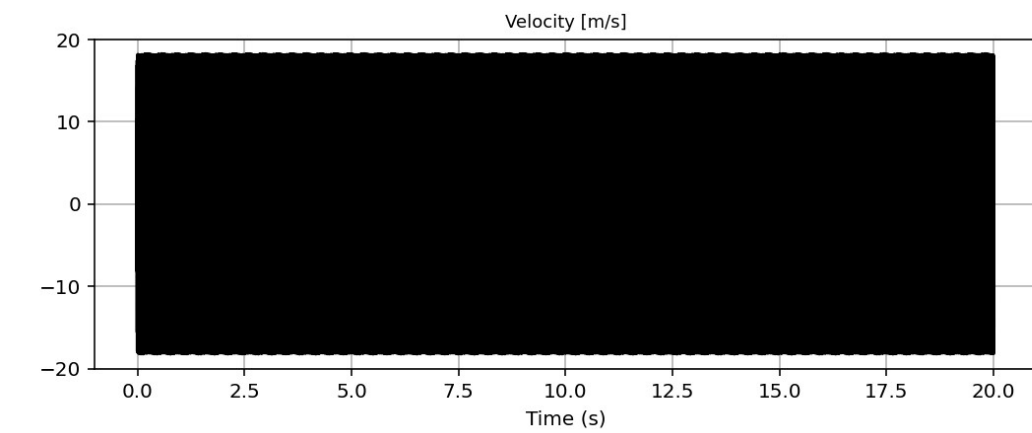
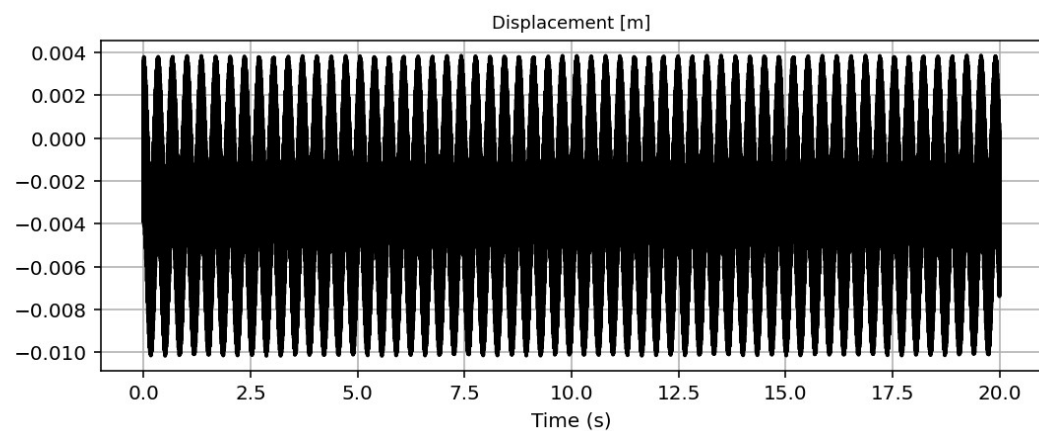
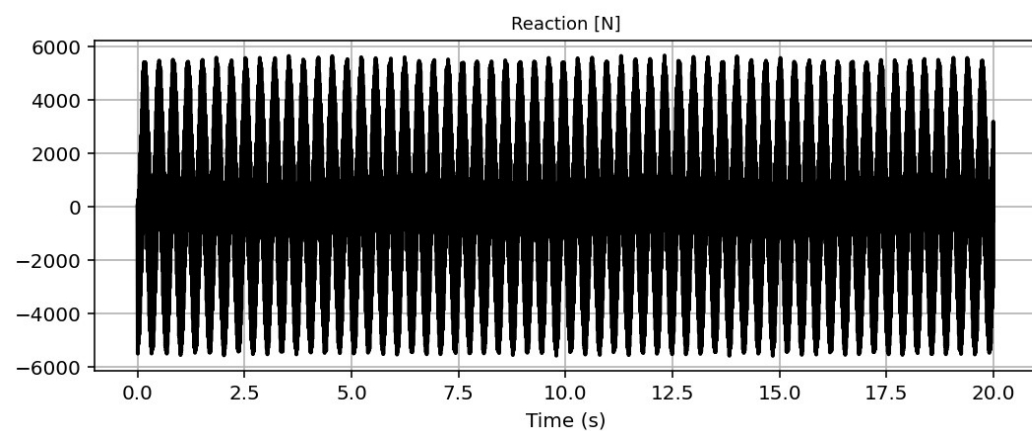
Base Reaction and Dispalcement of Structure During free-vibration Analysis

IPython Console Files Help Variable Explorer Debugger Plots History

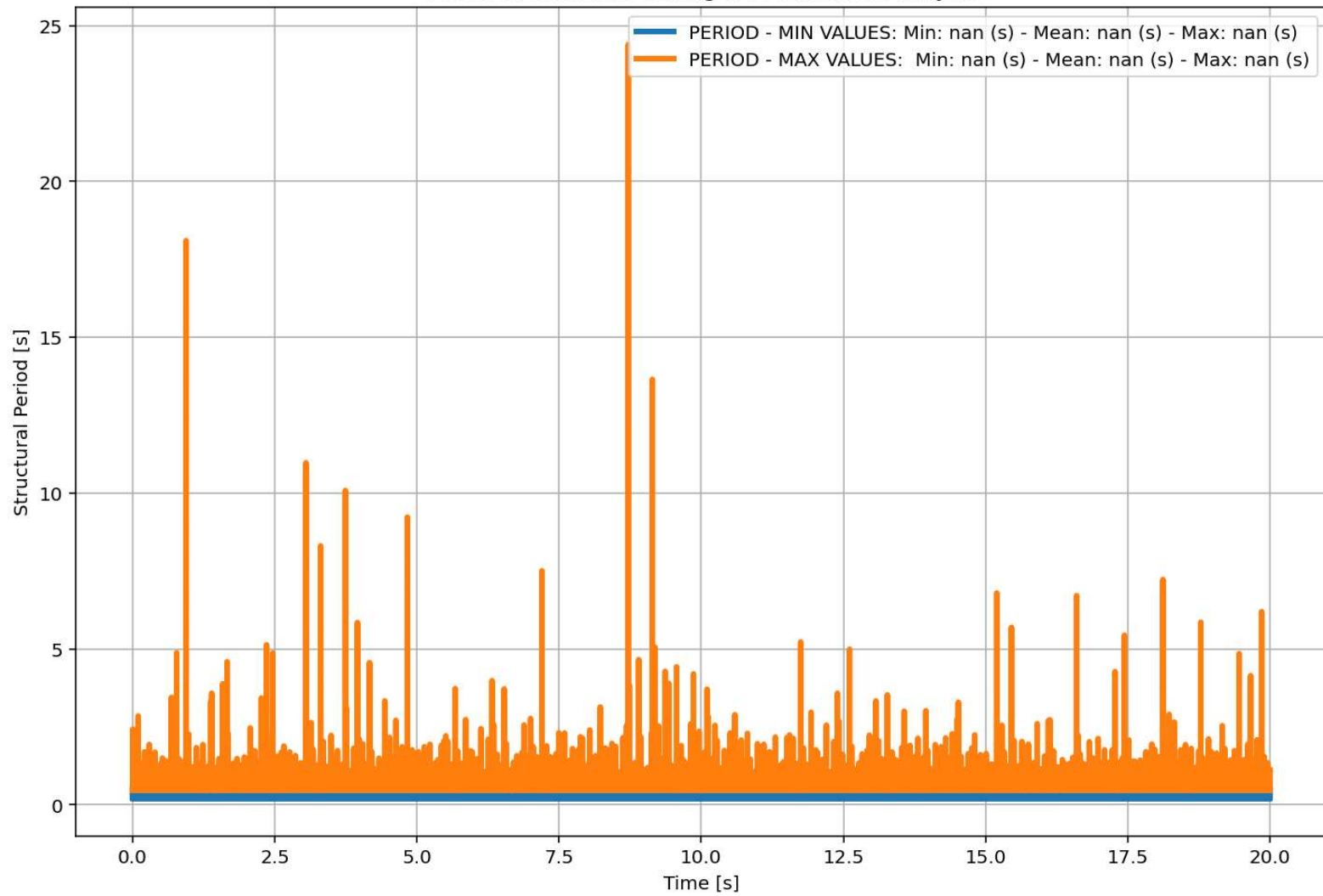
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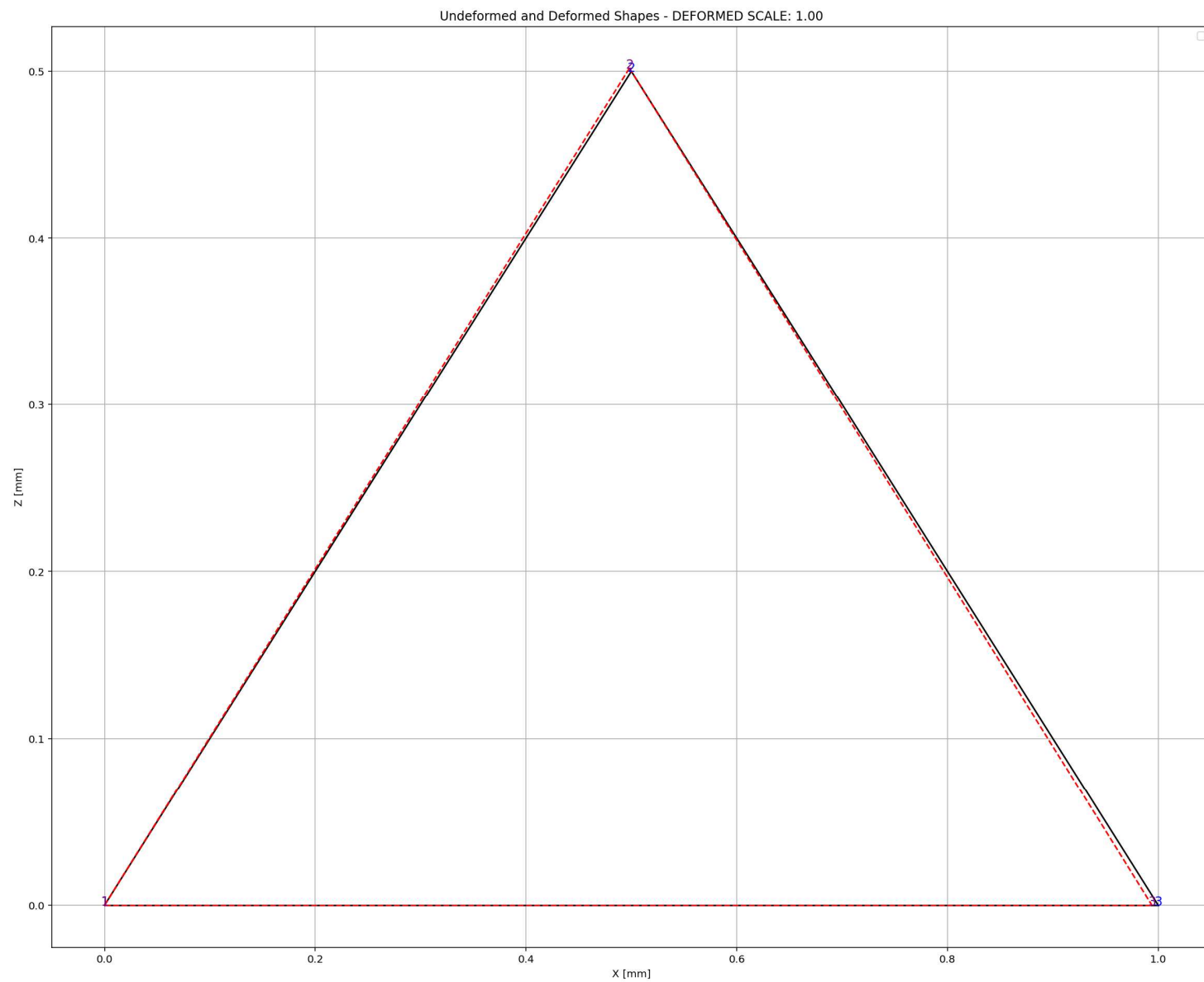
Base Reaction and Dispalcement of Structure During Free-vibration Analysis





Period of Structure During Free-vibration Analysis





SEISMIC ANALYSIS

Spdyer (Python 3.12)

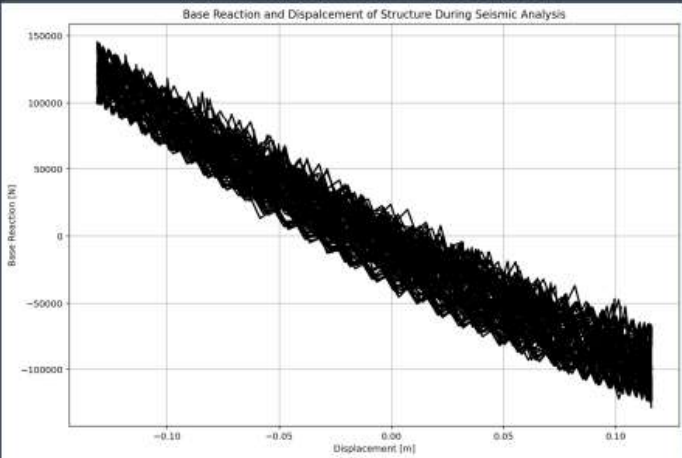
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..EX\25_FRAME_TRUSS_8_ANA\P(t)_FRAME_TRUSS_8_ANA.py

P(t)_FRAME_TRUSS_8_ANA.py X

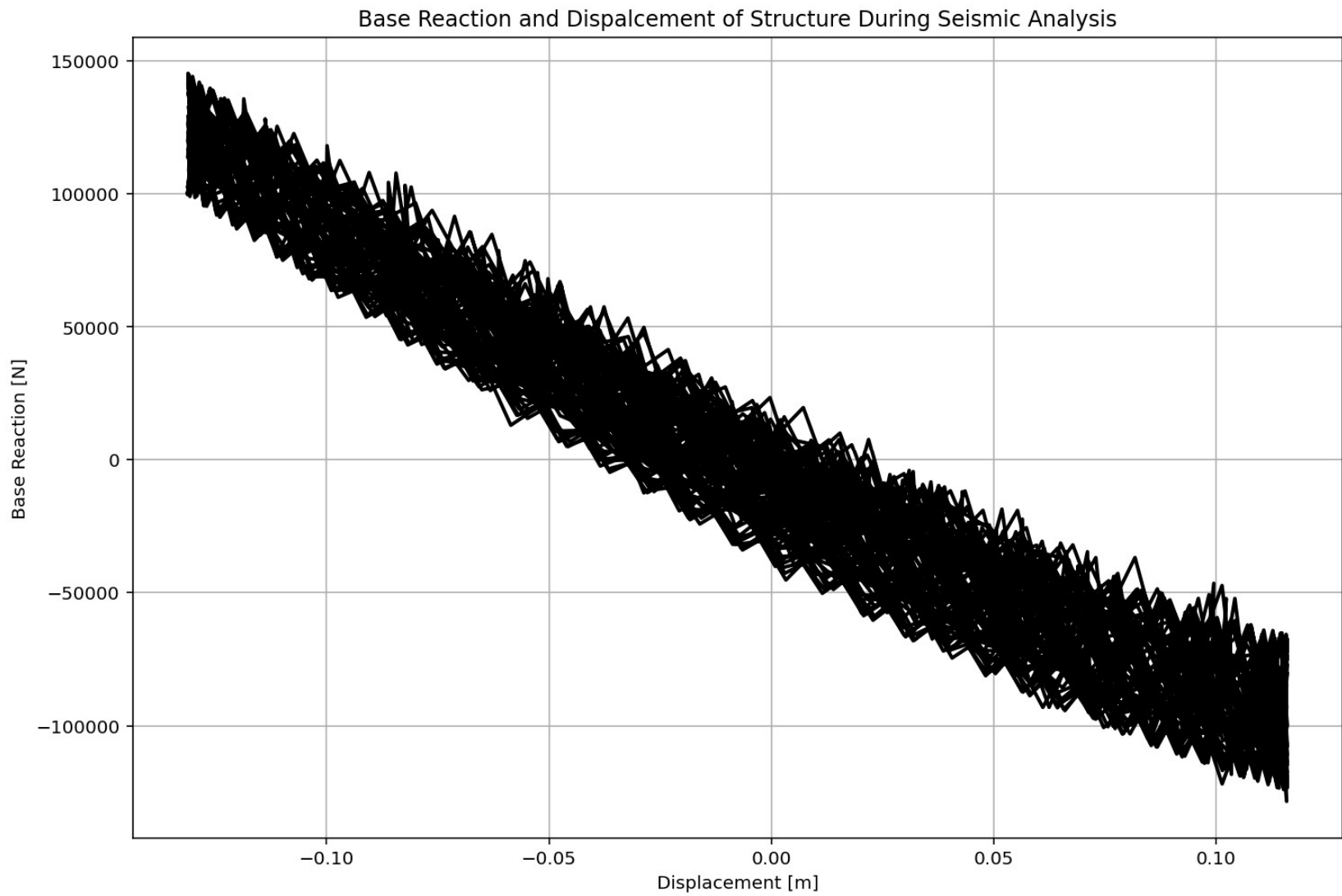
```
641 |
642 if ANAL TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
643     %% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
644     duration = 20.0           # [s] Analysis duration
645     dt = 0.01                # [s] Time step
646     GMfact = 9810            # [mm/s²] standard acceleration of gravity or st
647     SSF_X = 0.01             # Seismic Acceleration Scale Factor in X Direct
648     SSF_Y = 0.01             # Seismic Acceleration Scale Factor in Y Direct
649     iv0_X = 0.0005           # [mm/s] Initial velocity applied to the node
650     iv0_Y = 0.0005           # [mm/s] Initial velocity applied to the node
651     st_iv0 = 0.0             # [s] Initial velocity applied starting time
652     SEI = 'X'                # Seismic Direction
653     DR = 0.03                # Damping ratio
654
655 # Define time series for input motion (Acceleration time history)
656 if SEI == 'X':
657     SEISMIC_TAG_01 = 100
658     ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
659     # Define load patterns
660     # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
661     ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
662 if SEI == 'Y':
663     SEISMIC_TAG_02 = 200
664     ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
665     ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
666 if SEI == 'XY':
667     SEISMIC_TAG_01 = 100
668     ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
669     # Define load patterns
670     # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
671     ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
672     SEISMIC_TAG_02 = 200
673     ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
674     ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
```

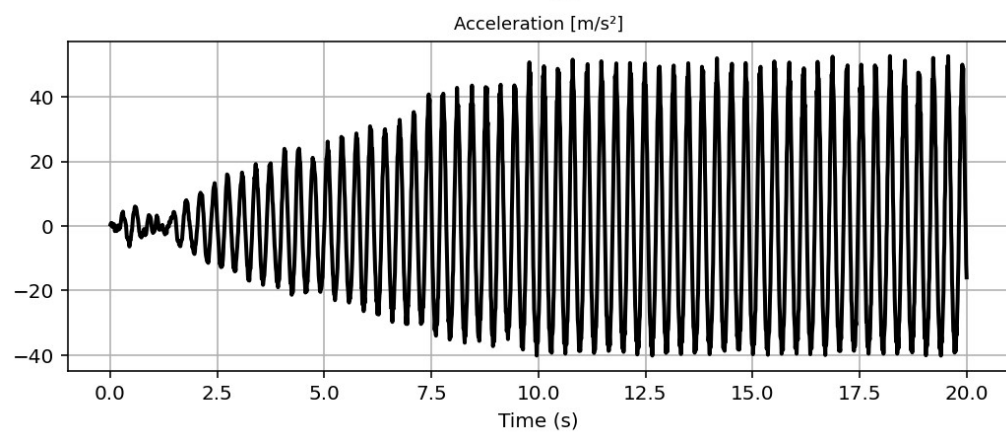
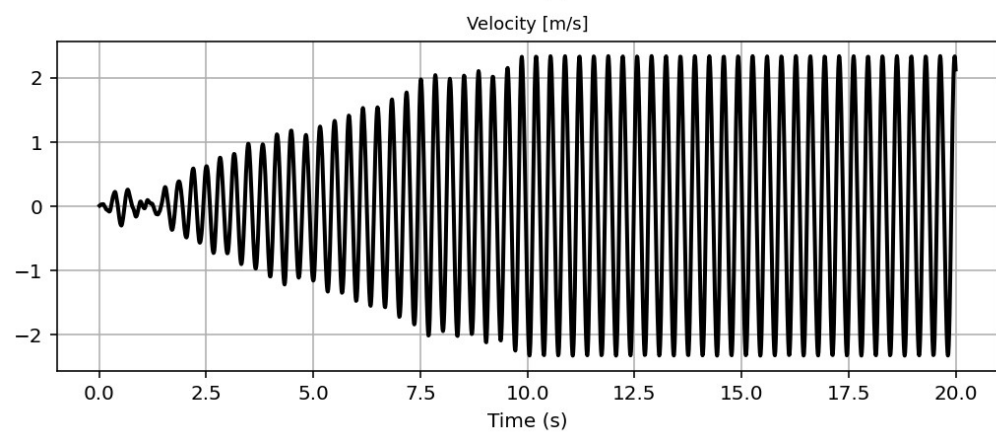
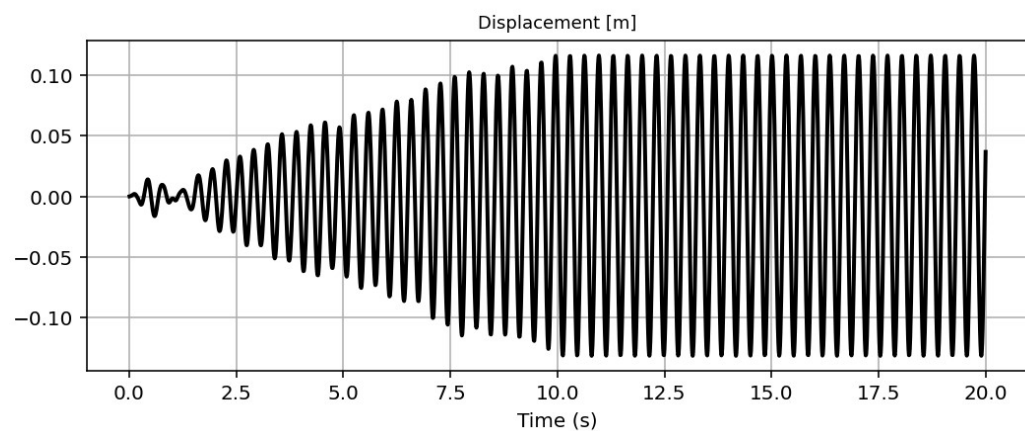
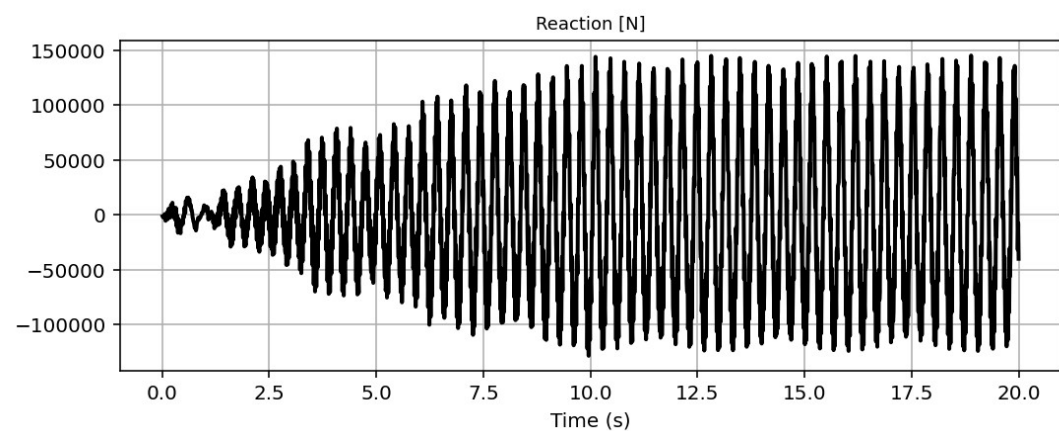
Base Reaction and Displacement of Structure During Seismic Analysis

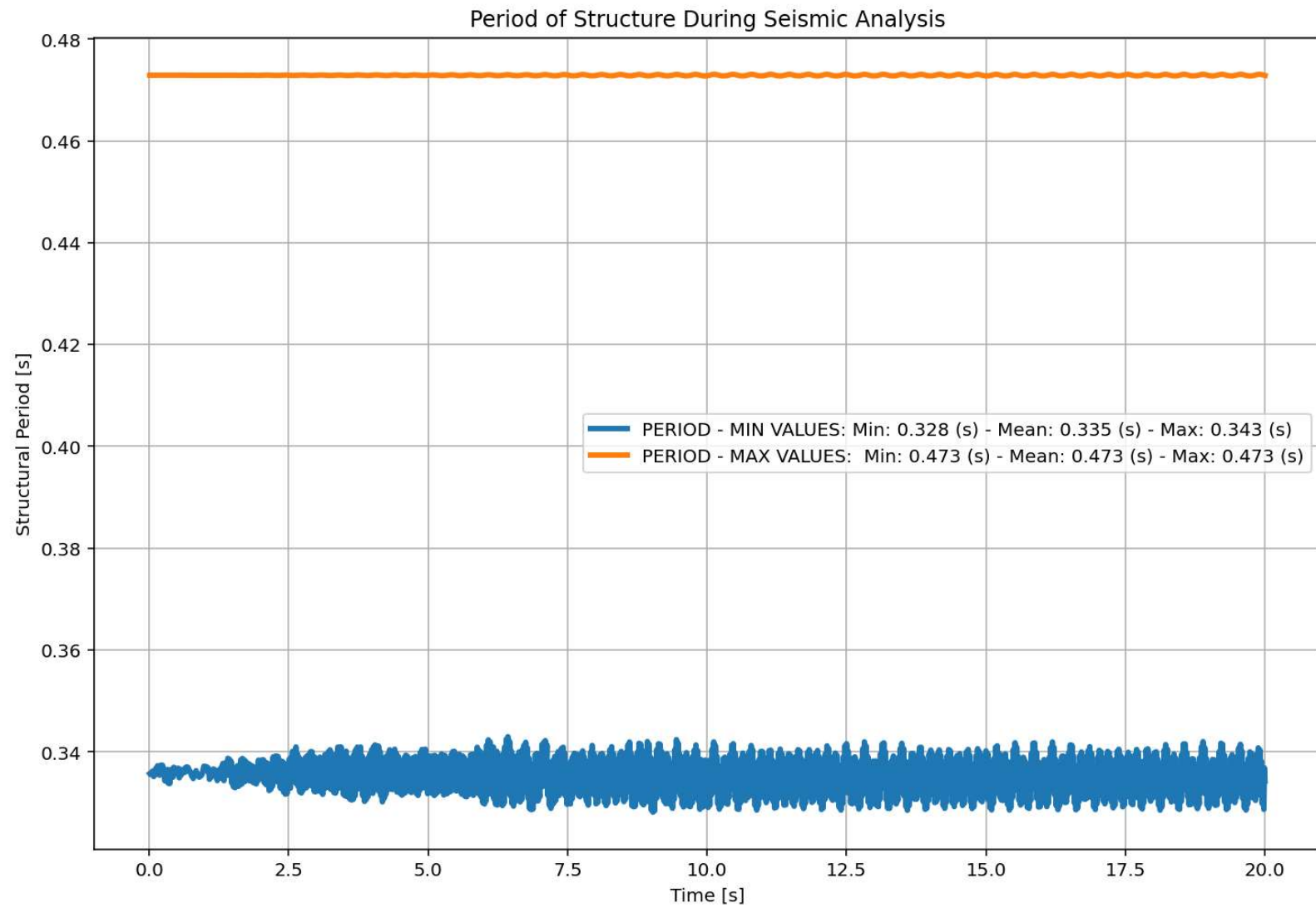


IPython Console Files Help Variable Explorer Debugger Plots History

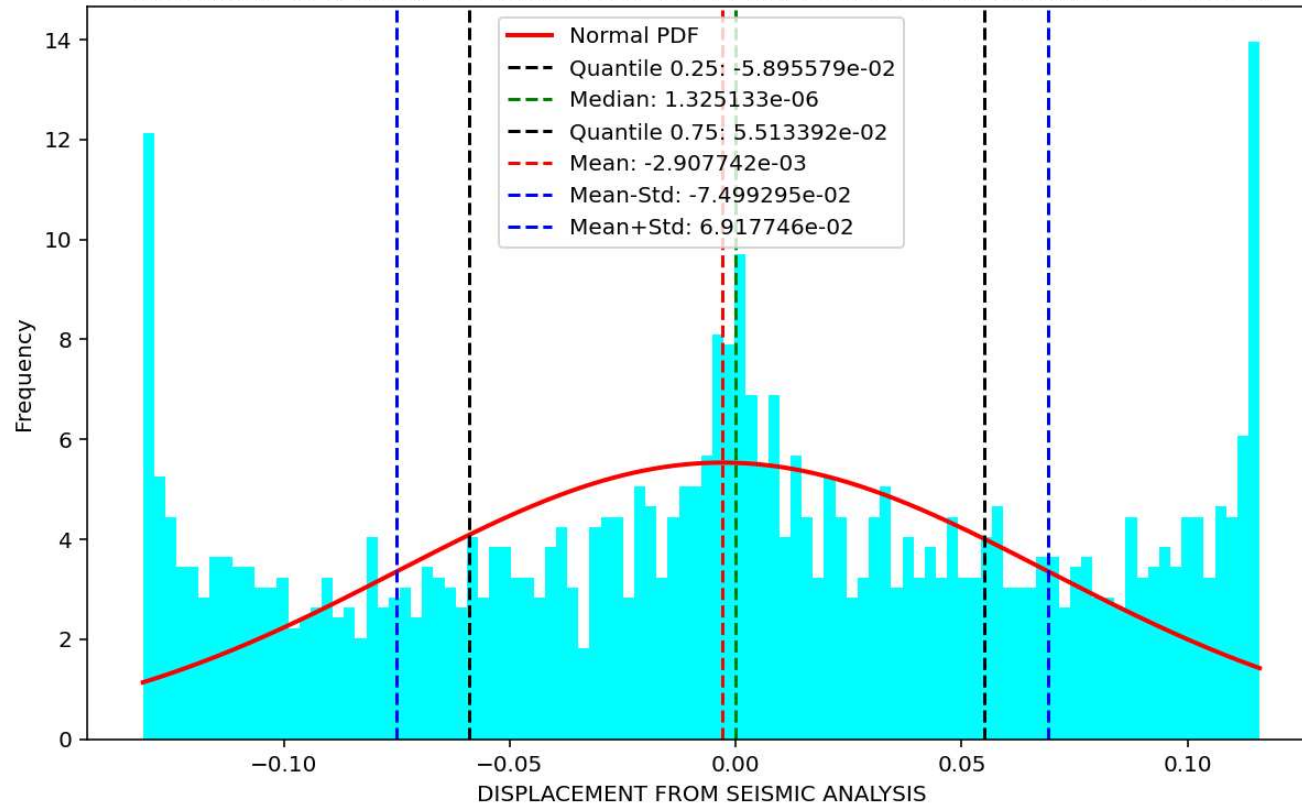
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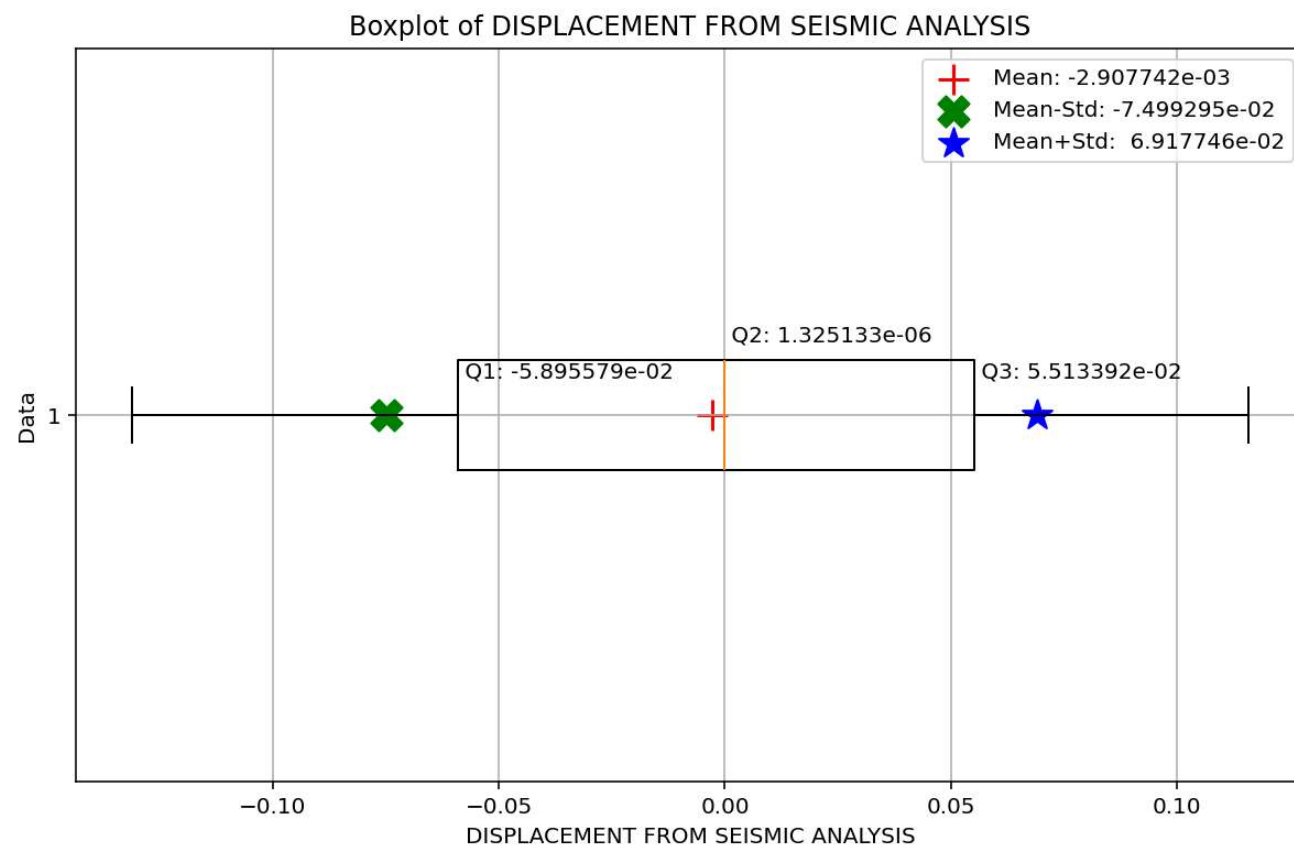


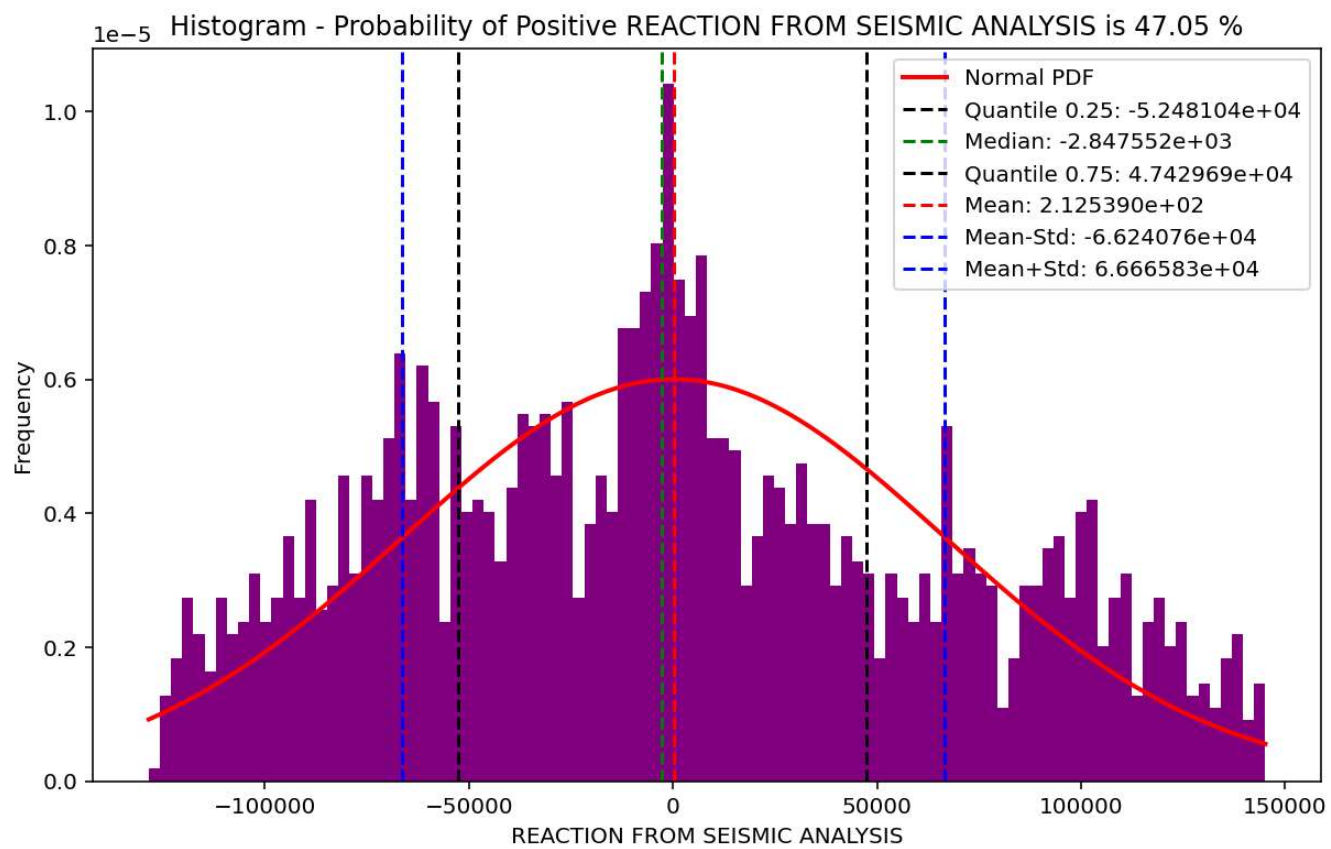




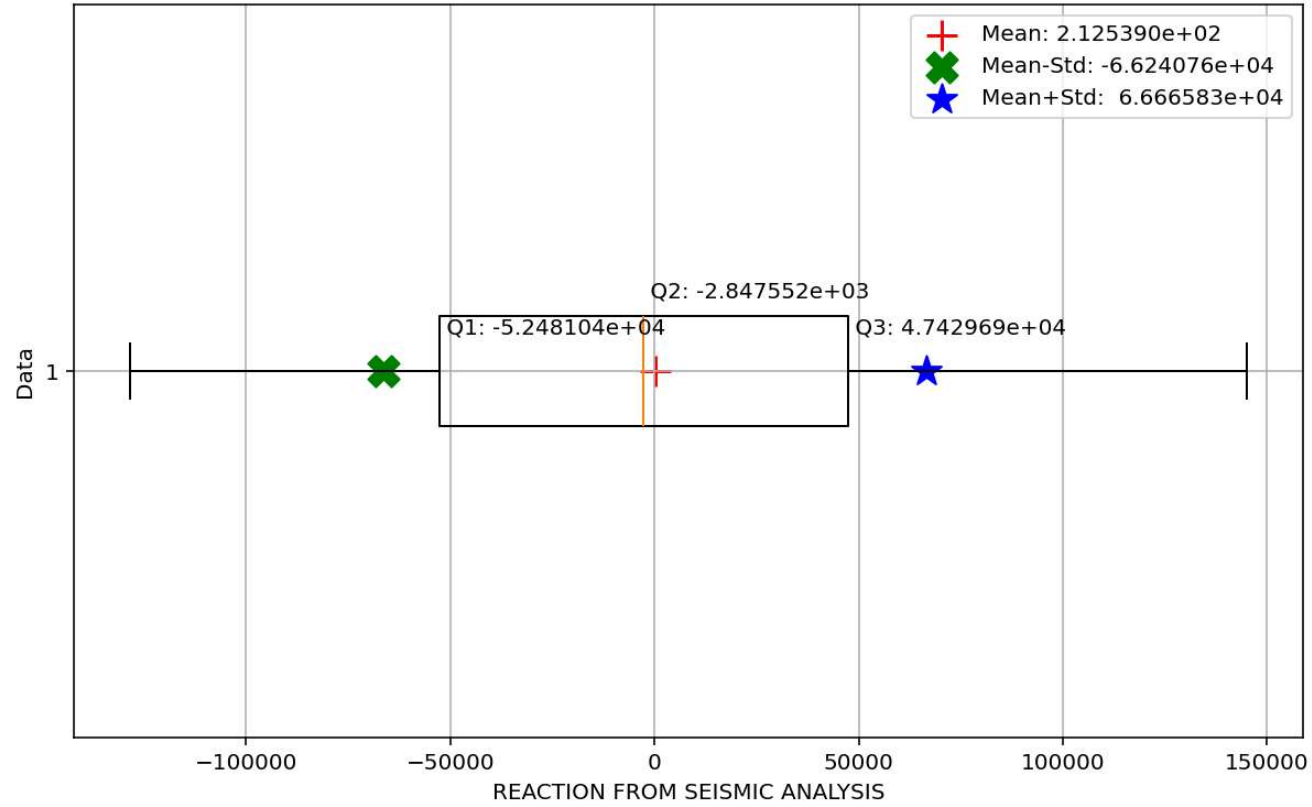
Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 50.00 %

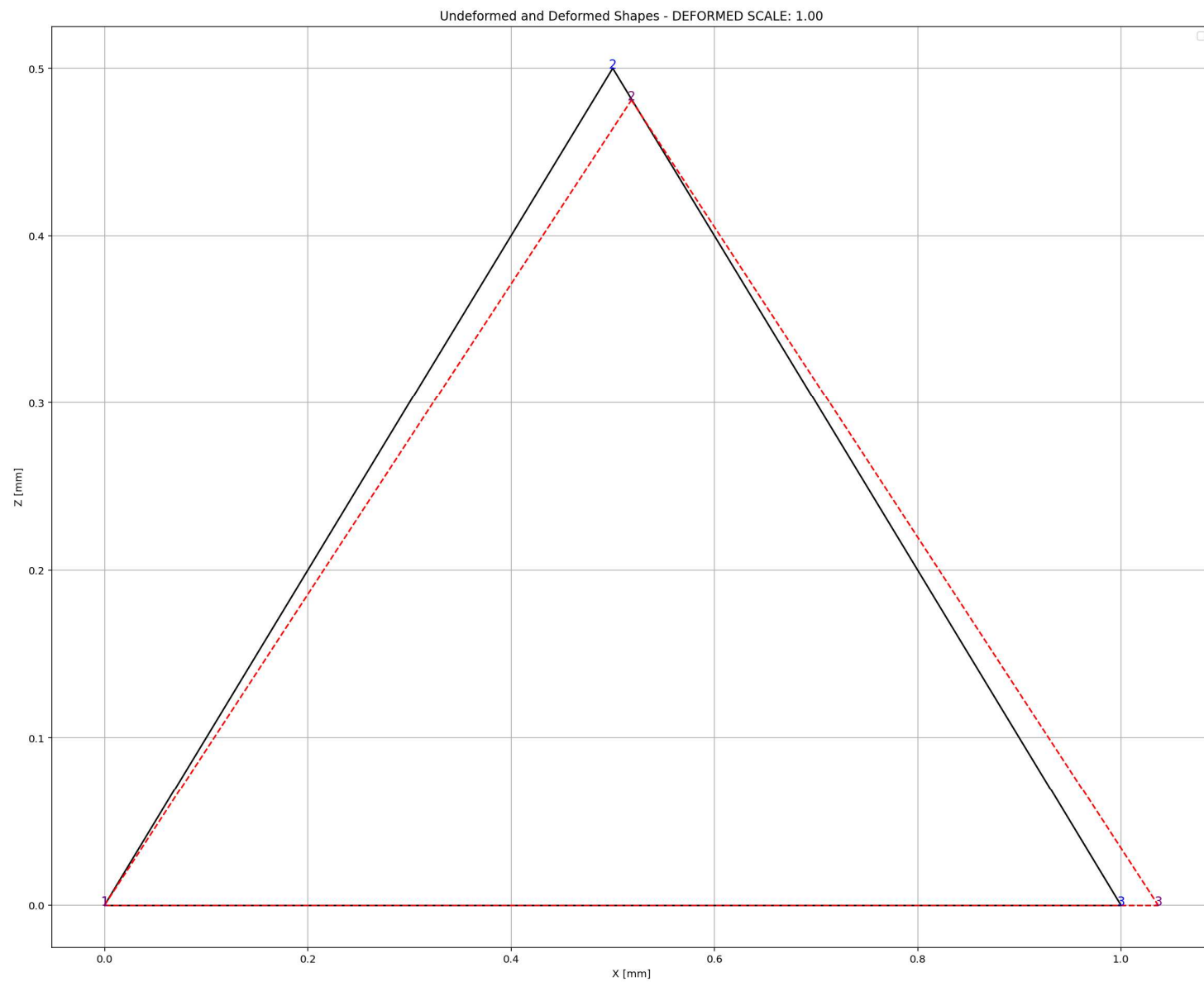




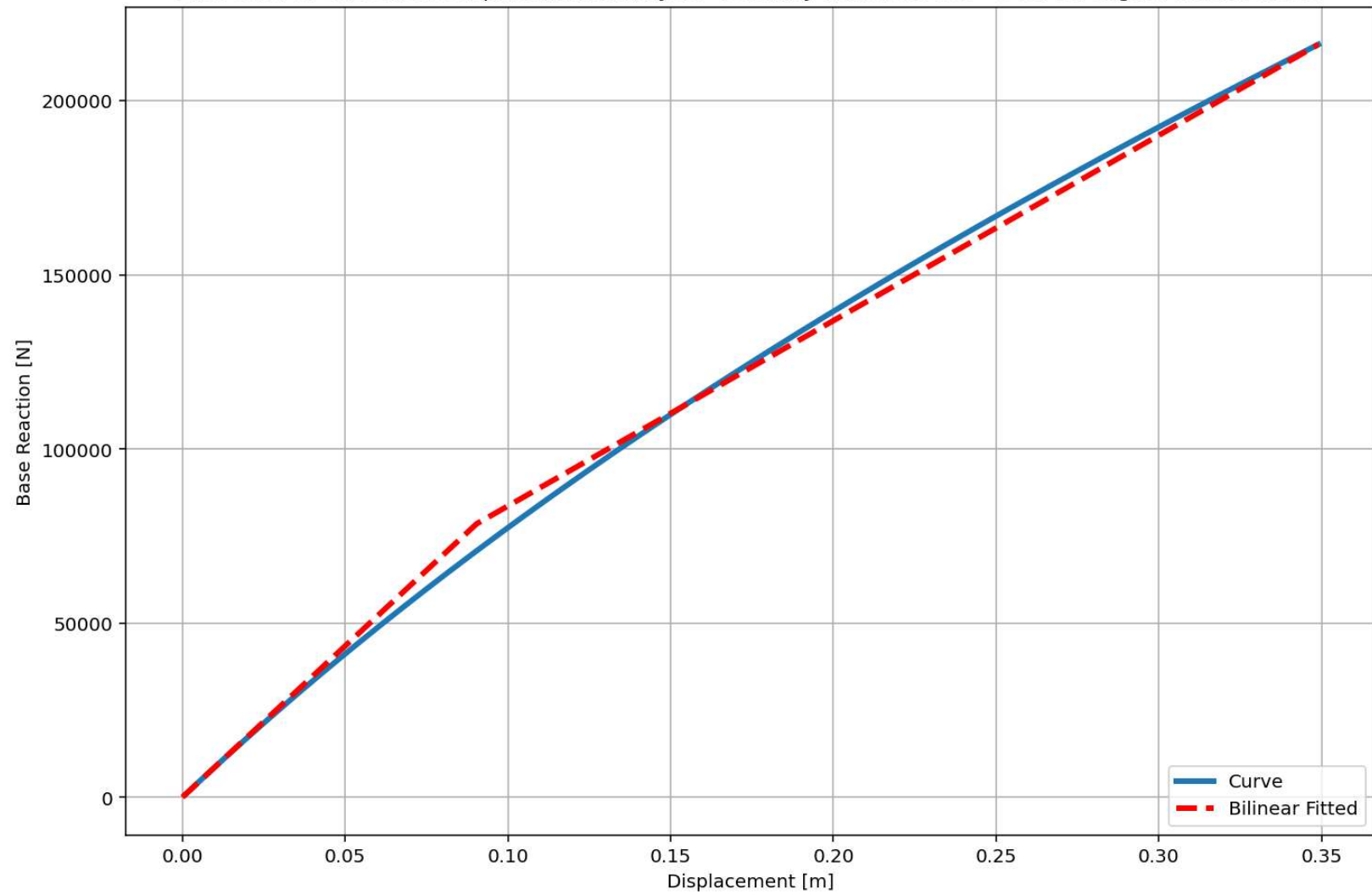


Boxplot of REACTION FROM SEISMIC ANALYSIS





Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 3.8625 - Over Strength Factor: 2.7539



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[9.03570300e-02 7.84853705e+04]

[3.49000000e-01 2.16143071e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 868613.88

Structure Plastic Stiffness : 619321.12

Structure Tangent Stiffness : 532230.59

Structure Ductility Ratio : 3.86

Structure Over Strength Factor: 2.75

+-----+

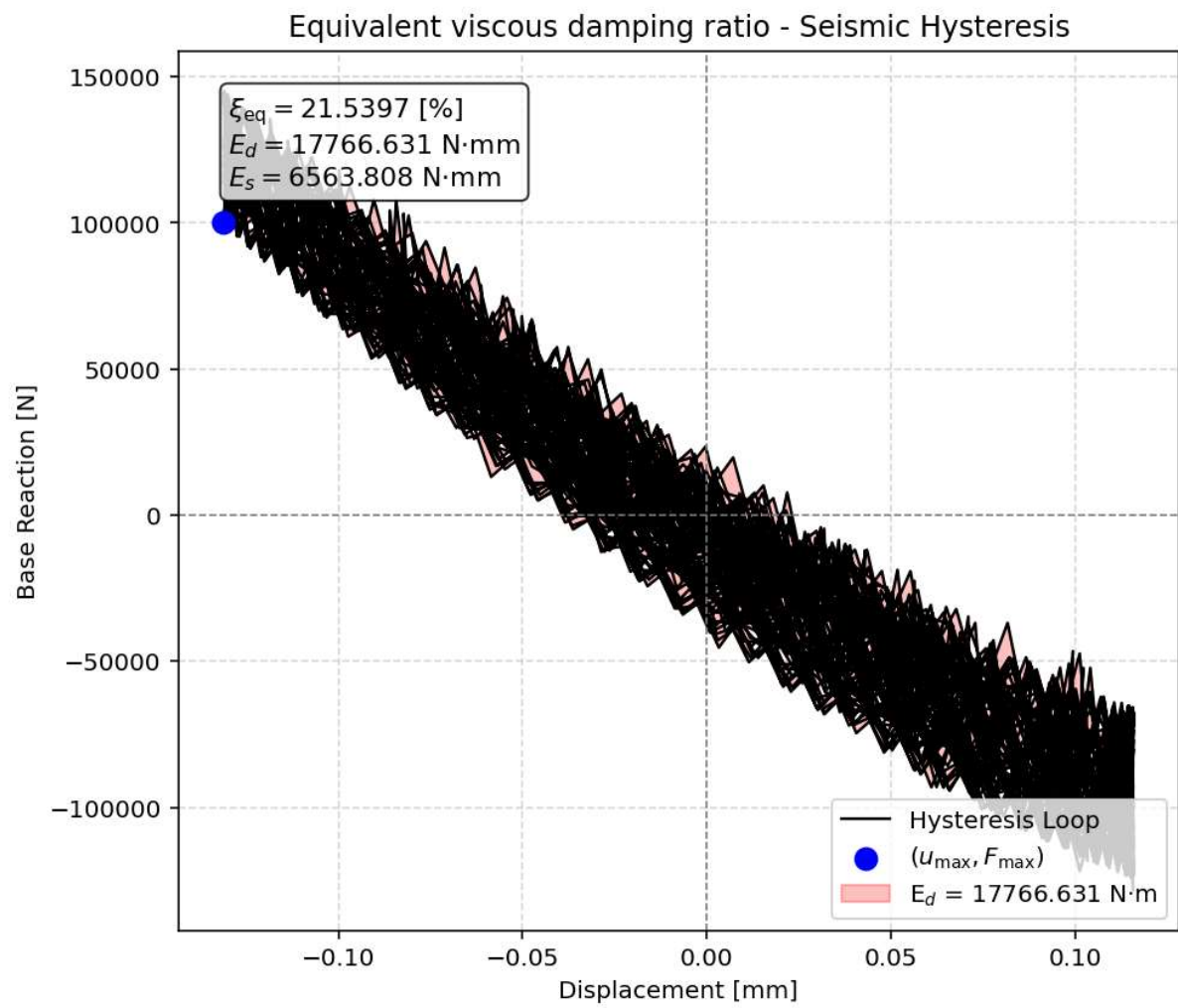
Over Strength Coefficient (Ω_0): 2.7539

Displacement Ductility Ratio (μ): 3.8625

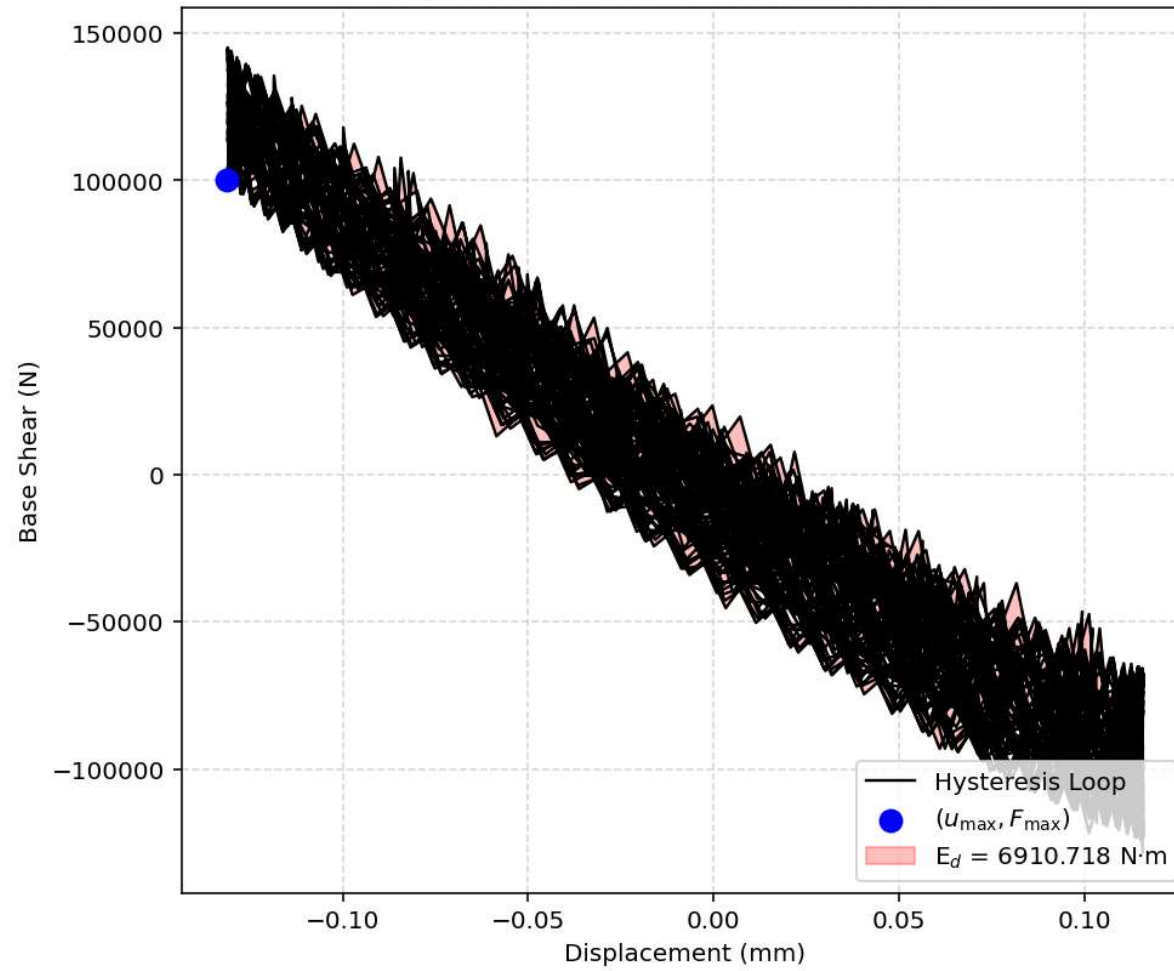
Ductility Coefficient (R_μ): 3.8625

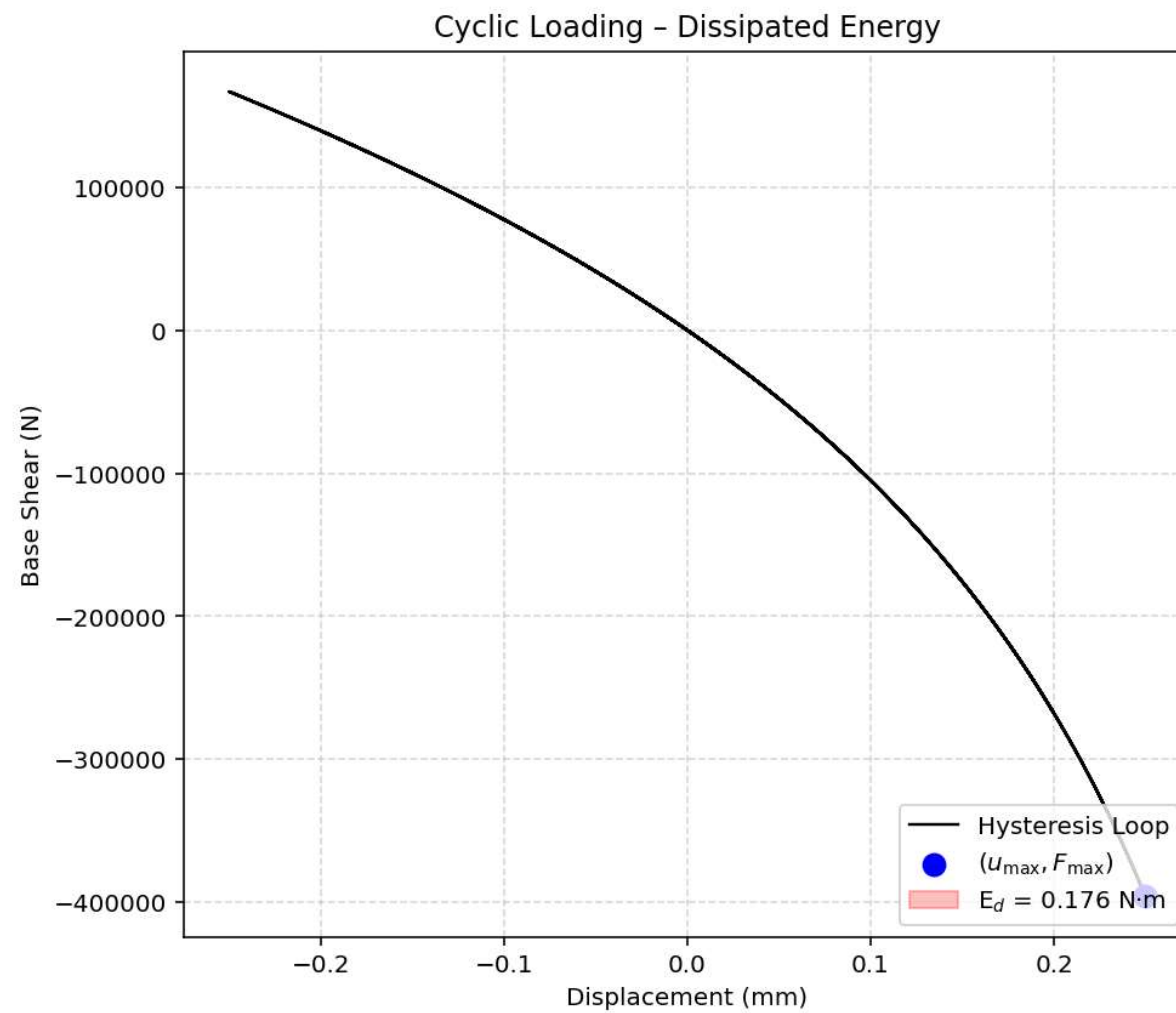
Structural Behavior Coefficient (R): 10.6369

Structural Ductility Damage Index in X Direction: 15.7960 (%)



Earthquake Response - Dissipated Energy





Dissipated Energy from Earthquake= 6910.72 N·mm
Dissipated Energy from Cyclic Displacement= 0.18 N·mm

**FOR EVALUATION OF DISSIPATED ENERGY CAPACITY INDEX:
CHECK THE CYCLIC DISPLACEMENT ANALYSIS AND IF IT IS POSSIBLE
INCREASE THE DISPLACEMENT.**

